

[illegible]

Manual

MYOCYTER v2.0 – Hardware and software

Please be aware that **ImageJ and FIJI** require lots of **RAM (16 GB+)** for processing of **large and/or high-resolution video material**.

MYOCYTER requires a **Windows-based operating system** like Win7/10/11.
Use of „**macOS**“ or „**OS X**“ may cause **dysfunction** of our software.

MYOCYTER v2.0 – Quickstart

This **quickstart guide** for **MYOCYTER 2** outlines its workflow as concisely as possible.

MYOCYTER is a **macro** for the **free image-processing software ImageJ/FIJ**. It enables the **extraction of a comprehensive range of parameters from videos of contracting cells**, such as (cardio-)myocytes.

Please read the **complete quickstart guide first** to familiarize yourself with the software's capabilities and to determine **which settings are best suited** to your samples and data.

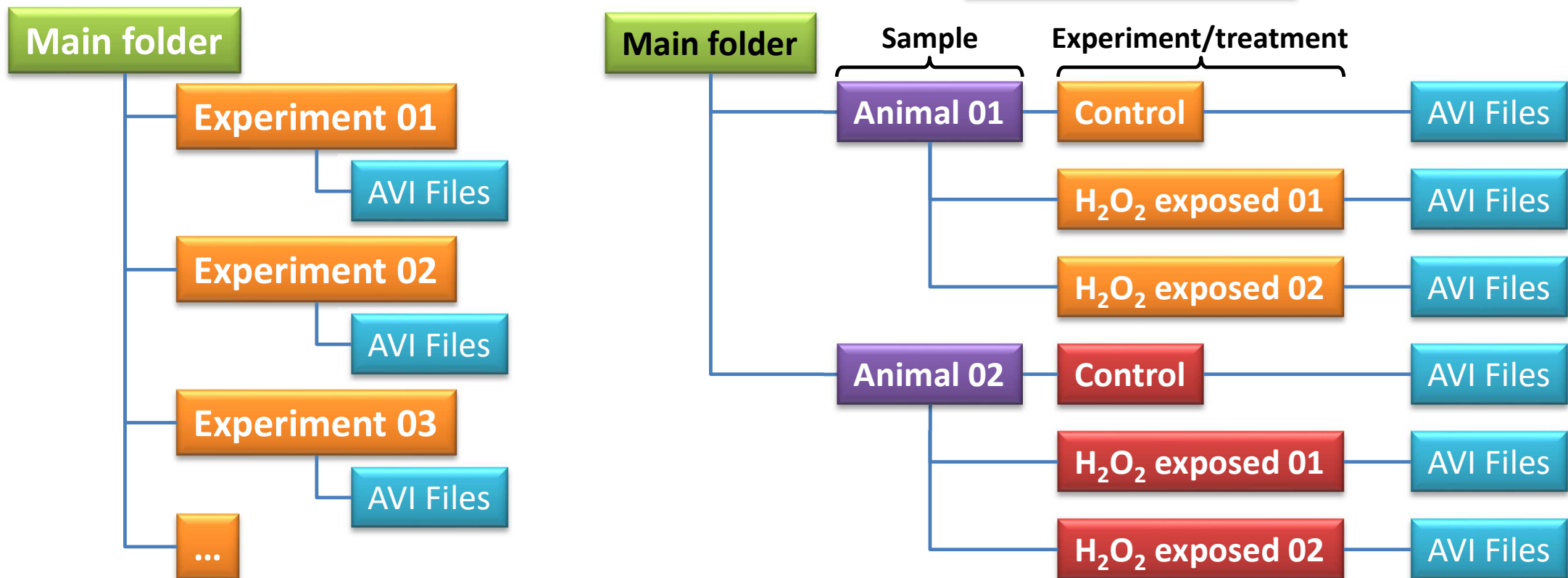
The latest version of MYOCYTER, the quickstart guide, the detailed manual, and several test videos can be found at: www.scyrus.de

You may also visit our [YouTube channel](#) or discuss the macro on [GitHub](#).

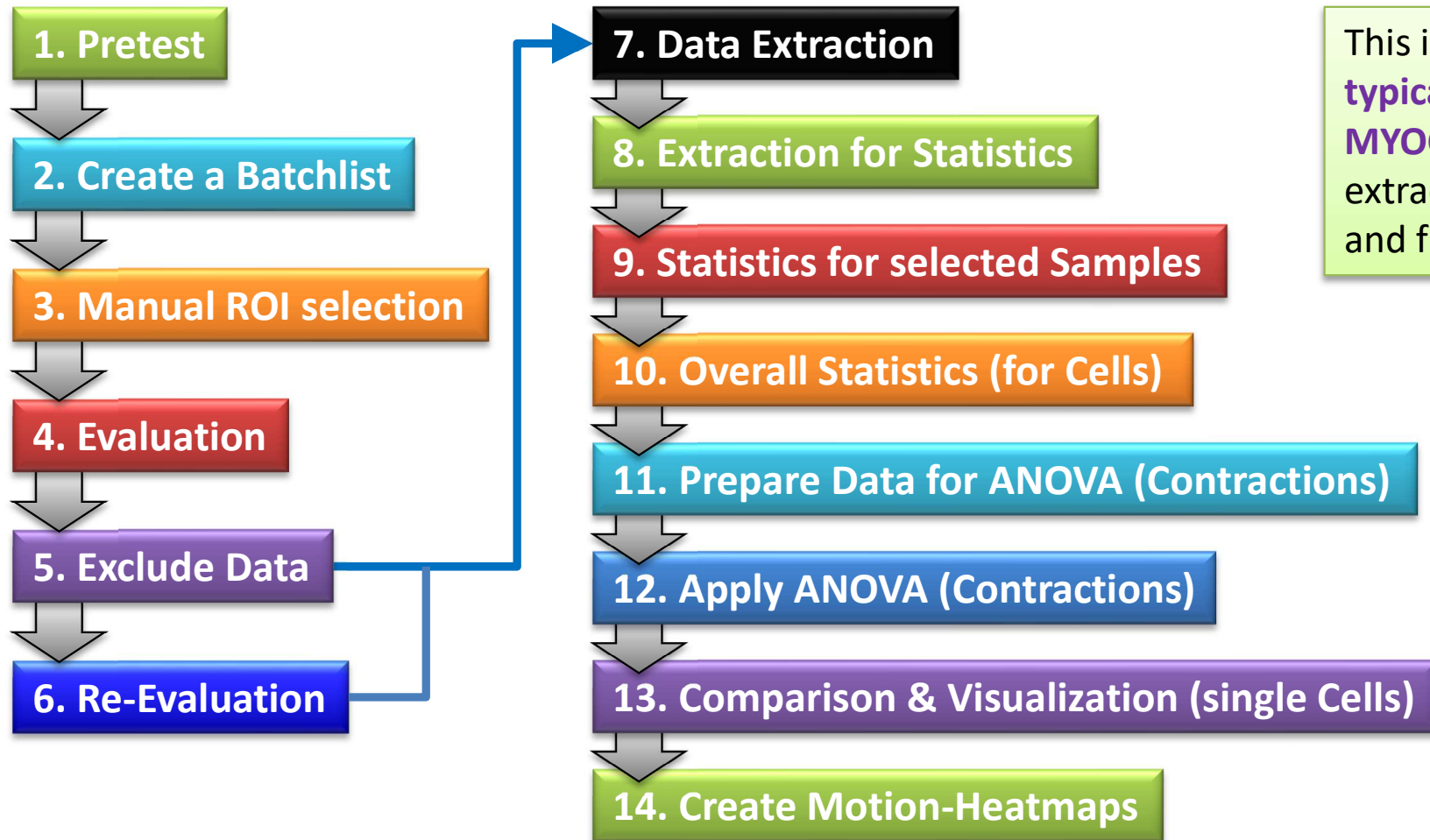
How to arrange your data for MYOCYTER v2.0

“**Experiment**” refers to cell recordings from the **same petri dish** (i.e. from the **same animal** and/or under the **same treatment**).

Realistic example



The Workflow of MYOCYTER v2.0

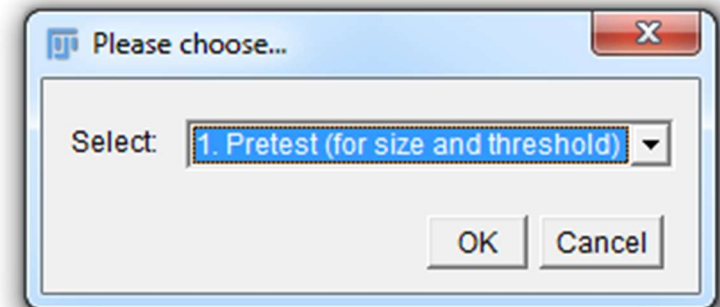


This is both the **menu** and the **typical workflow of MYOCYTER v2.0** during data extraction from videos and further processing.

The Workflow of MYOCYTER v2.0

0. Before you start

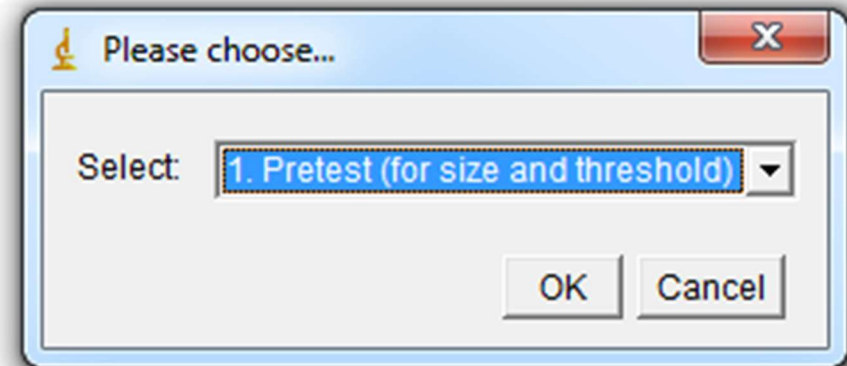
- You have all your videos in an **ImageJ-compatible format** (**MJPEG** or **uncompressed *.avi** files) in a folder.
- Use a current version of FIJI ([Link](#)).
- You need the **MYOCYTER-Macro** as *.txt file.
- A computer with **≥16 GB of RAM** and a **64-bit Windows-OS** is recommended.
- Open the file containing the macro code, copy it and **start ImageJ**.
- Select „Plugins“ → „Macros“ → „Record“.
- Paste the code into the new window („**Recorder**“), click the „**Create**“-button and close the „Recorder“-window.
- The macro code is now in a new window („**Macro.ijm**“). Select „Macros“ → „Run Macro“ (or press „Strg+R“).
- MYOCYTER starts and you should see its **user-interface**.



The Workflow of MYOCYTER v2.0

1. Pretest

- The **pretest** assures **optimal recognition** of contracting cells and **exclusion of the background**.
- Select the first option „**1. Pretest (for size and threshold)**“ from the user interface and click „OK“.

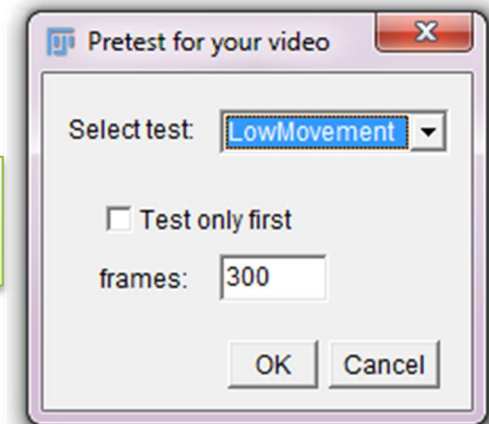


The Workflow of MYOCYTER v2.0

1. Pretest

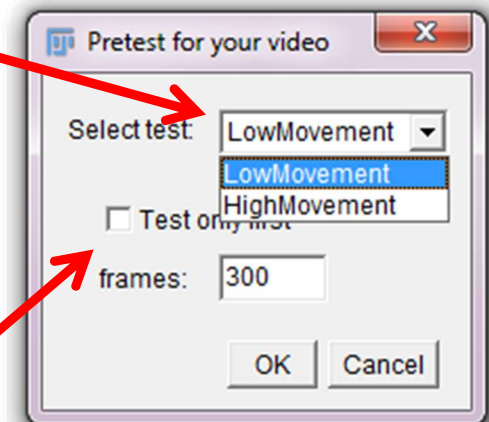
- Stick with the **default settings** (for the beginning).
- **Select** the folder containing your videos, **DO NOT open** it, click „**OK**“ and **wait** until the macro has finished.
- A new folder („**pretest**“) should be found in the folder containing your videos.

Default settings



Use „LowMovement“ for cells that show little contraction – which setting to use comes with some experience.

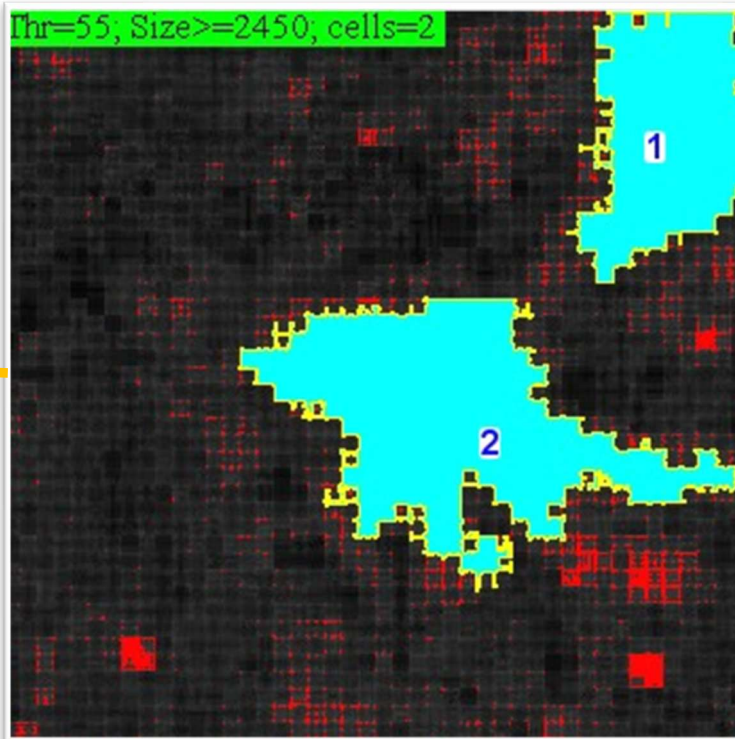
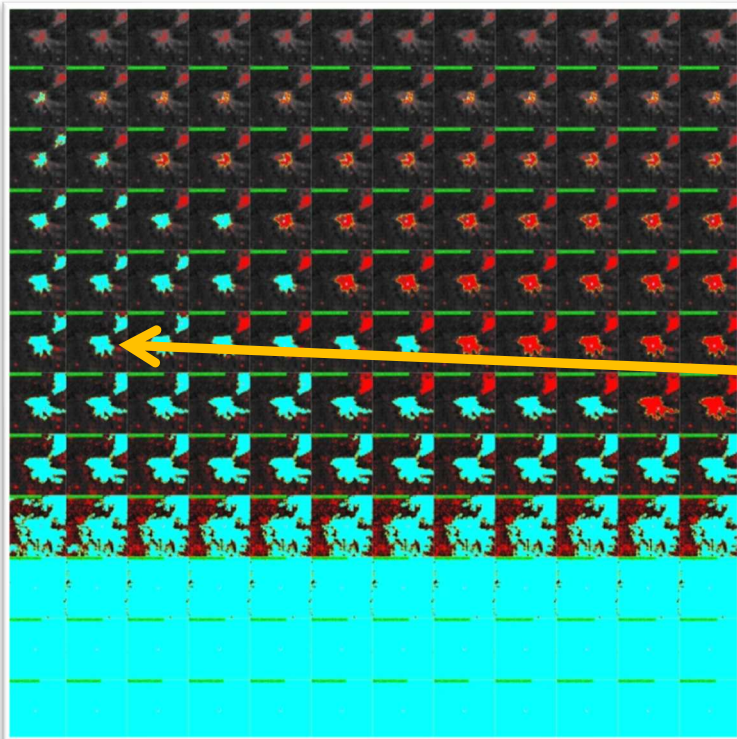
To save time, pretest only the first xxx frames of your video



The Workflow of MYOCYTER v2.0

1. Pretest

The **images of the pretest** should look like this:

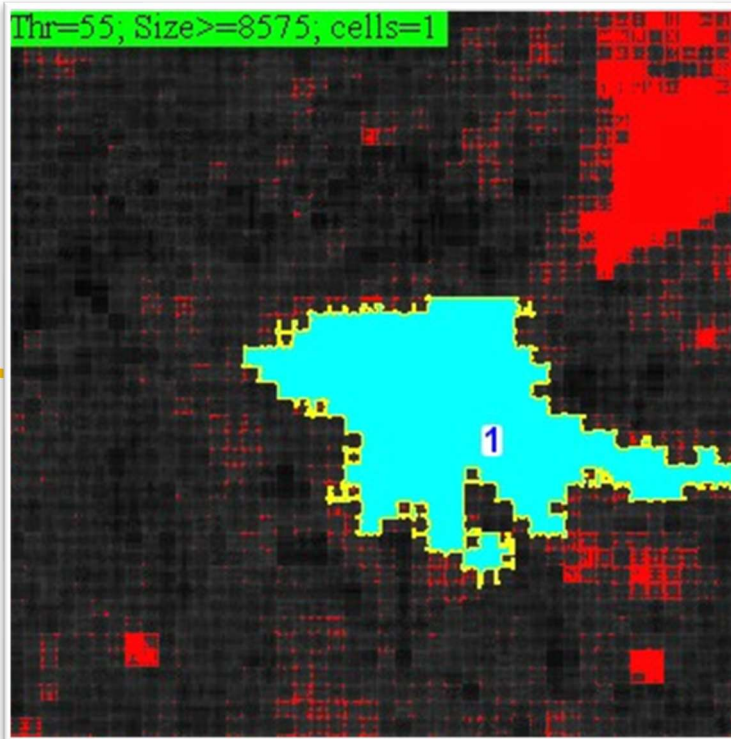
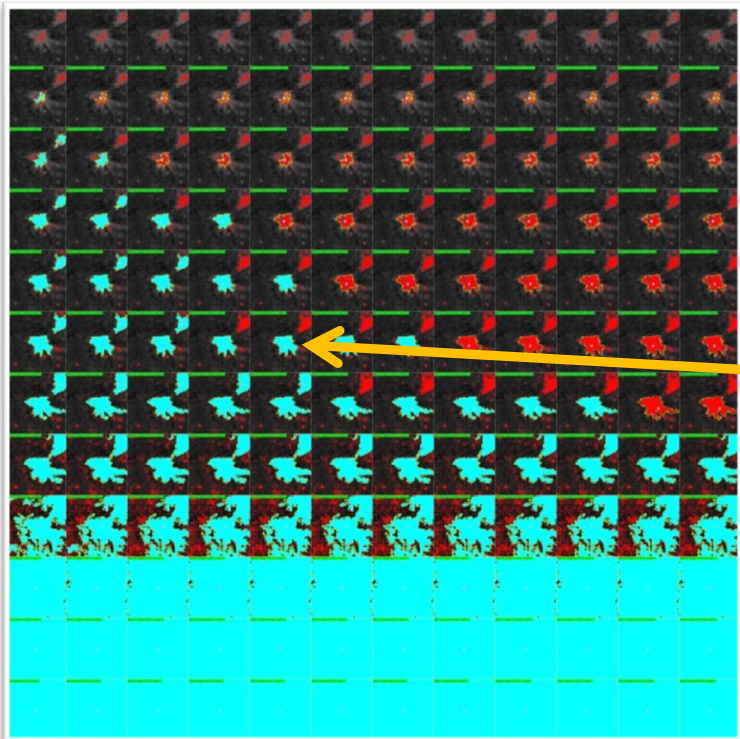


Two cells have been recognized. To restrict evaluation to the moving cells only, we should apply (in this case) a **Threshold of 55** and a **Size of 2450**. **Only the turquoise areas** will be evaluated with the corresponding settings, while the **background is completely excluded**.

The Workflow of MYOCYTER v2.0

1. Pretest

The **images of the pretest** should look like this:

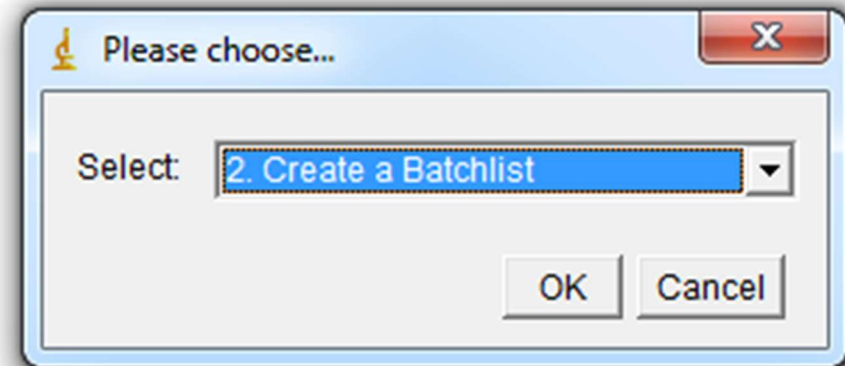


Two cells have been recognized.
To exclude the one cropped through the picture frame, apply a **Threshold of 55** and a **Size of 8575**.
Only the turquoise areas will be evaluated with the corresponding settings, while the **background is completely excluded**.

The Workflow of MYOCYTER v2.0

2. Create a Batchlist

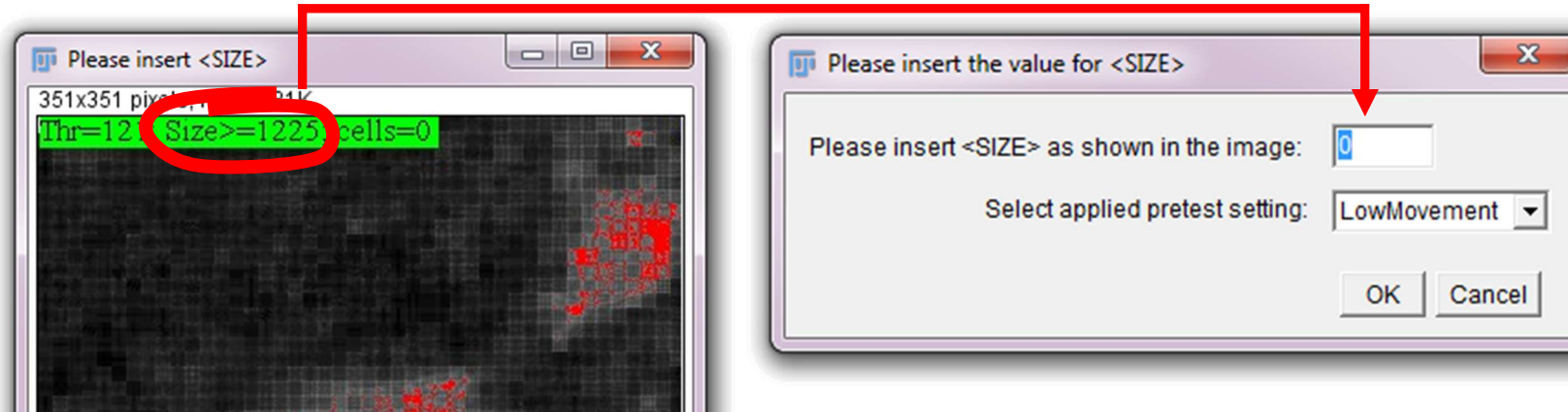
- Application of **individual settings** for **Threshold** and **Size** from „1. Pretest“ **to every single video** can be completely automatized.
- Select option „**2. Create a Batchlist**“.
- **Just select** the folder containing your pretest images, **DO NOT** open it.
- Click „**OK**“.



The Workflow of MYOCYTER v2.0

2. Create a Batchlist

- An image opens and you will be asked to insert the „**Size**“ value (left image) into the **corresponding field** (right image).



Please type the value for „**Size**“ in this field (in this case 1225). This has to be done **only once**.

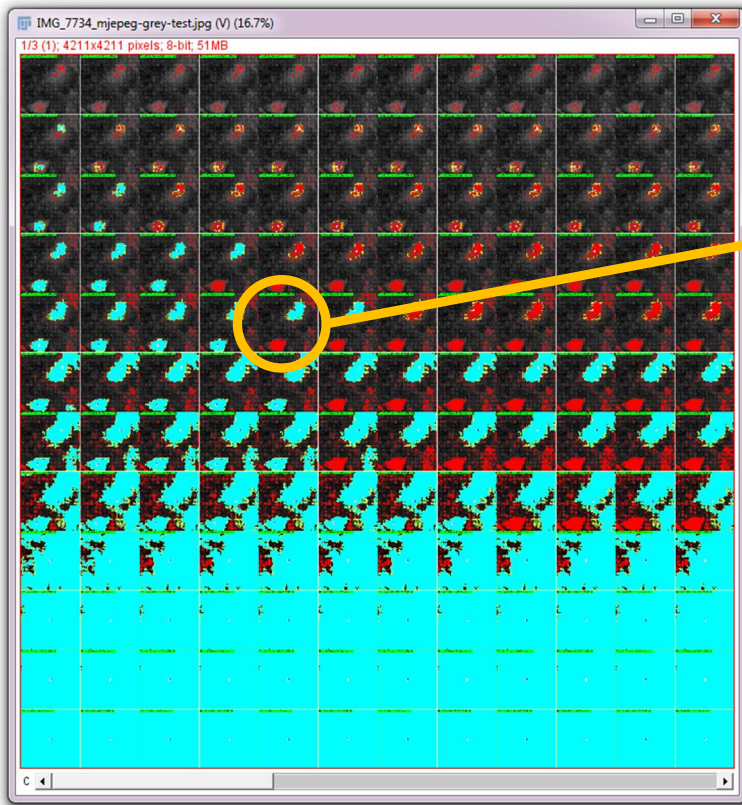
NOTE: All videos in the folder should have the same dimensions!

- After this, please select „**LowMovement**“ or „**HighMovement**“ – depending on your **pretest settings**. This information is also given in the **name of the folder** containing the pretest images. For example: „pretest (**LowMovement**) 2019-Sep-21 22h09“.
- Click „**OK**“ to start selection of the individual settings.

The Workflow of MYOCYTER v2.0

2. Create a Batchlist

- Now, **just left click the panel** of the pretest images **matching the settings you want to apply**.



Just **left click** the **desired panel** (small images).

To **magnify the image**, you can use the **mouse wheel** while pressing „**Strg**“.

After selection, the current pretest image will be closed and the next one opens automatically until every single one has been processed.

The Workflow of MYOCYTER v2.0

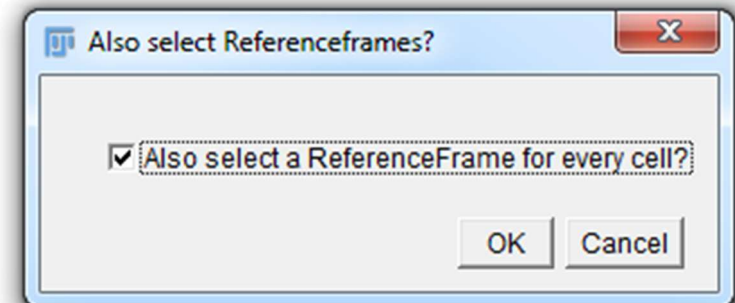
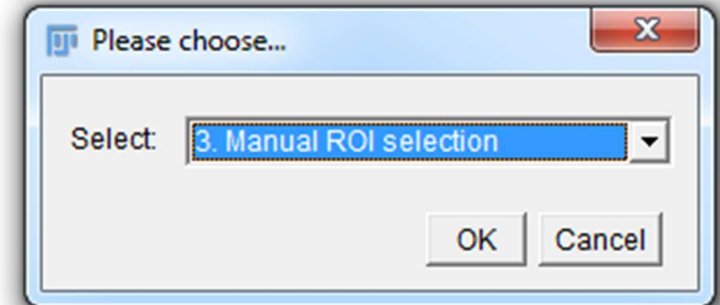
2. Create a Batchlist

- After processing the last pretest image, the **individual settings** will be found as table (**batch.txt**) in the pretest folder.
- **Copy this file (batch.txt)** into the **folder containing your videos**.
- Using the file **batch.txt**, MYOCYTER applies individual settings for every single video during evaluation. **ONLY** the videos **actually listed in this file** will be processed.

The Workflow of MYOCYTER v2.0

3. Manual ROI selection

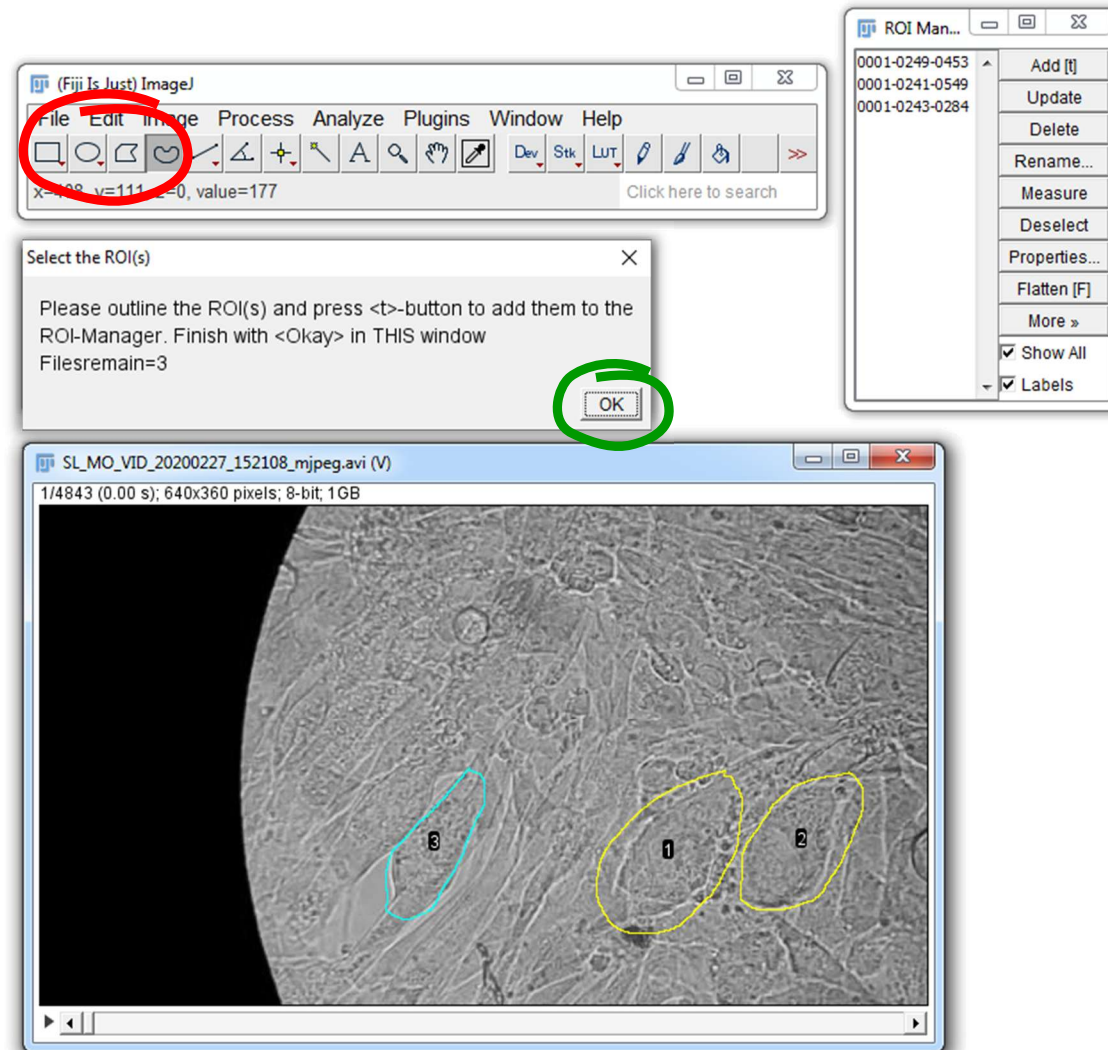
- Select „**3. Manual ROI selection**“ and click „**OK**“.
- This feature enables **manual outlining** and **labelling of cells without any pretest**.
- Just select (**do not open**) the folder containing your videos, click „**OK**“.
- In the next step you will be **asked about (manual) selection of a reference frame** for every single defined cell. In most cases, you can uncheck this feature, because **automatic recognition is quite reliable and robust**.



The Workflow of MYOCYTER v2.0

3. Manual ROI selection

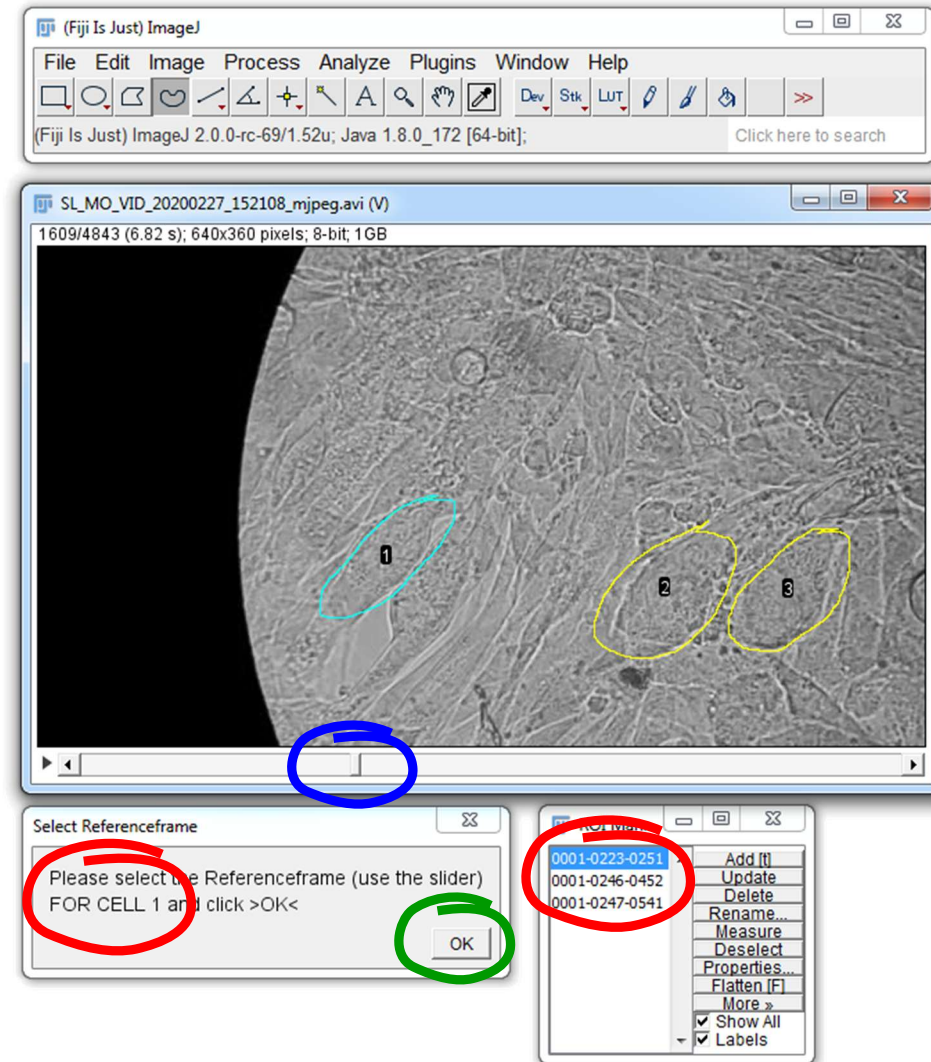
- Now, please select (**do not open**) the folder containing your video-files.
- Every single file will be opened successively and the user is asked to **outline single cells**.
- Pick one of the **selection tools** (**red circle**).
- **Label the first cell** (see video-panel) and press the „t“-key to assign the selection to the „**ROI Manager**“ (top right panel). Repeat for **every single cell** to be analyzed.
- To **finish the current video**, click „**OK**“ in the „**Select the ROI(s)**“-window (**green circle**).



The Workflow of MYOCYTER v2.0

3. Manual ROI selection

- IF you checked „Also select a ReferenceFrame for every cell“ at the beginning, you are asked to adjust the according reference frame for every single selection using the slider of the video (**blue circle**) after defining the individual ROIs.
- Define the **appropriate reference frame** for every single ROI (**red circles**) using the slider of the video frame, then click „OK“ (**green circle**) to proceed to the next ROI.



The Workflow of MYOCYTER v2.0

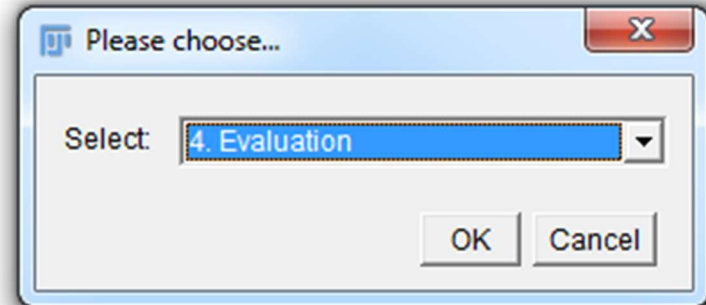
3. Manual ROI selection

- Repeat this selection of ROIs and also reference frames (if you activated this option) for **every single video**.
- MYOCYTER will **terminate after the last file**, and a „manrois“-subfolder is created in the folder containing your videos.
- How this manual ROI selection is used during evaluation is explained in the next section (4. Evaluation).

The Workflow of MYOCYTER v2.0

4. Evaluation

- Select „4. Evaluation“ and click „OK“.



The Workflow of MYOCYTER v2.0

4. Evaluation

This creates an animated plot of every single recognized cell synchronized with its amplitude. This takes time, so you may apply this feature only to selected videos.

Higher values will recognize smaller amplitudes as contractions. **Can be changed even after evaluation.**

If checked, you will be asked to define the **resting state** of every single cell manually. Usually, **automatic recognition** of the **reference-frame** works very fine, so this field can be left unchecked.

Activate „**Batching**“ if you want to use a **Batch List**; the values for „Lower Threshold“ and „Size“ can be ignored and are overwritten by the individual ones of the list.

Please insert values for lower t...

Lower Threshold: 22

Size: 1600

☐ Video output

[%] of max recognized as beat: 20

Detection: 4

☐ Smoother

☐ Force reference-frame

☒ Replace zero from RefFrame

☒ Batching

☐ Use manual ROI-list

OK Cancel

If you have **only a single video** or videos with **very similar cells**, you can apply pretest-values for **Threshold** and **Size** directly here without creating a batch-list and **deselect „Batching“**.
If you want to **evaluate the whole frame** without restriction to the moving cells only, apply a **Threshold of 0** (Size becomes irrelevant in this case) and **deselect „Batching“**.

If the **amplitude** exceeds this value (here: 20%) of its **local peak**, the times for **systole** and **diastole** are measured. The values **10%, 50% and 90%** are applied anyway. This value **can also be changed even after evaluation**.

Check this option to use the **manual ROI-selection** created in section „3. Manual ROI selection“.

The Workflow of MYOCYTER v2.0

4. Evaluation

- After applying the desired **settings for evaluation**, click „**OK**“.
- **Just select** the folder **containing your videos** (and also the file „**batch.txt**“, if you selected „**Batching**“), **DO NOT** open it.
- Evaluation **may take some time**, depending on file size and number of your videos. After finishing evaluation ALL windows will be closed automatically.



Please **DO NOT** use the computer for **any other task during evaluation** – this **may lead to premature termination** of MYOCYTER.

Please insert values for lower t...

Lower Threshold: 22

Size: 1600

☐ Video output

[%] of max recognized as beat: 20

Detection: 4

☐ Smoother

☐ Force reference-frame

☒ Replace zero from RefFrame

☒ Batching

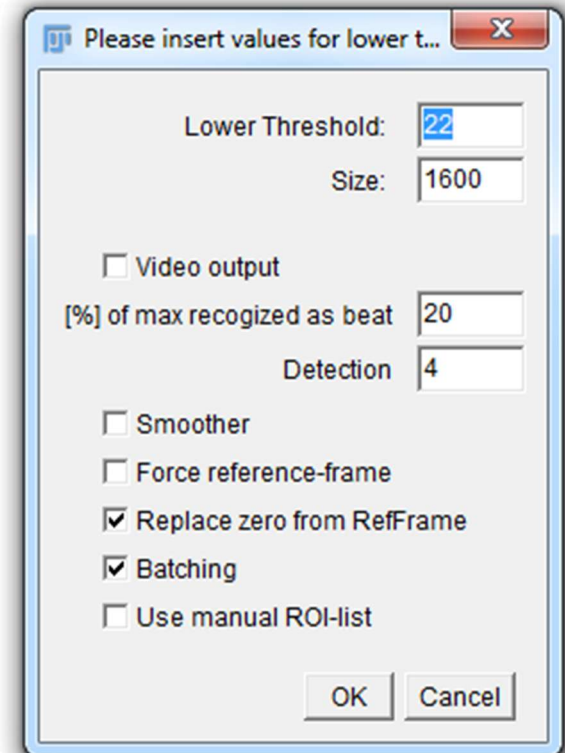
☐ Use manual ROI-list

OK Cancel

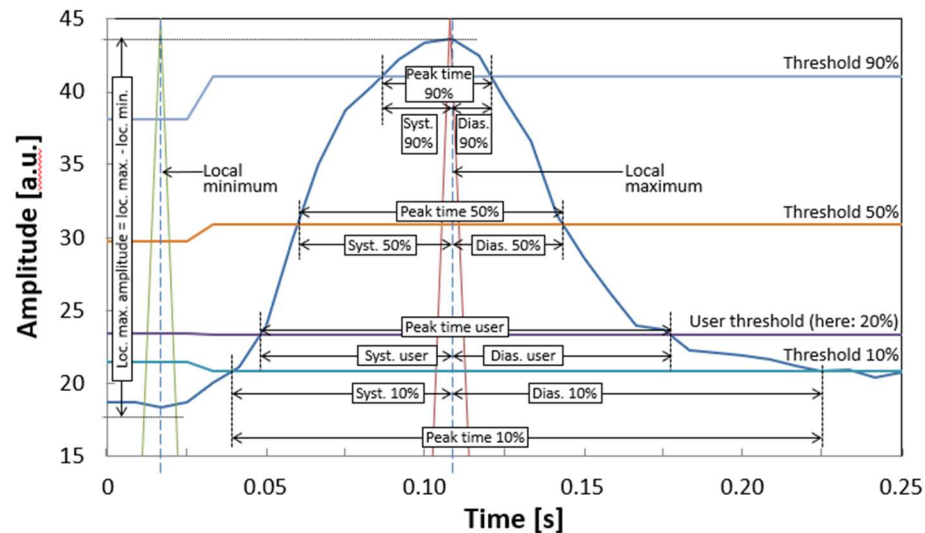
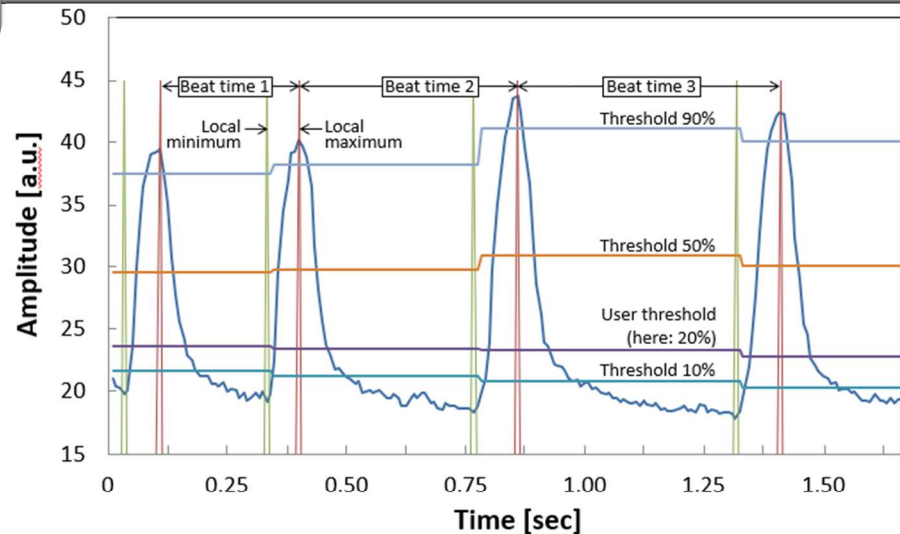
The Workflow of MYOCYTER v2.0

4. Evaluation

- If you want to **apply manual ROIs** (see section 3.), please check the option „**Use manual ROI-list**“.
- First, you may see a „warning“ that „batching“ was deactivated. This is just a hint that **batch-lists will be ignored**, click „OK“ to go on.
- First, you are asked to select the folder containing your videos (**just select, do not open it**).
- Second, you are asked to select (**not open**) the „**manrois**“-**folder** containing the ROI-data (please see section 3 for details).



The extracted data



The **data extracted** from the videos include, among other parameters, the **contraction and relaxation times** at the **four thresholds** (10%, 20%, 50%, and 90%, horizontal lines), the corresponding **peak times** as the sum of contraction and relaxation times, the **local maximum** (vertical dashed line), the **intervals between successive contractions**, the **amplitude** (dark blue curve), as well as **local minima** (green vertical spikes) and **local maxima** (red vertical spikes).

The Workflow of MYOCYTER v2.0

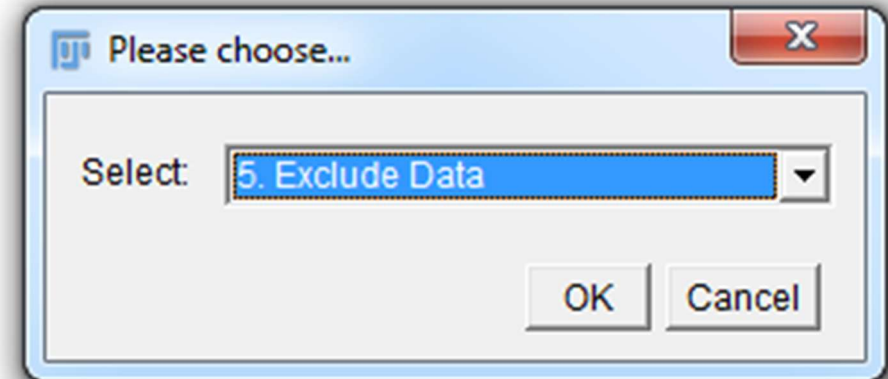
5. Exclude Data

- After evaluation, new **folders are created** in the one containing your videos: This folder should have the **subfolders „diffMov“ → „dataplots“**.
- The **folder „dataplots“** contains (amongst other things) **plots of the amplitude of every single recognized cell** as well as a **text file named „Results“**.
- It's possible that cells were evaluated that the **user wants to exclude from further analysis** - nobody wants to do this manually in a table that contains many thousands of lines.
- This process (**excluding data**) is largely automatized, the **original data remain untouched**.

The Workflow of MYOCYTER v2.0

5. Exclude Data

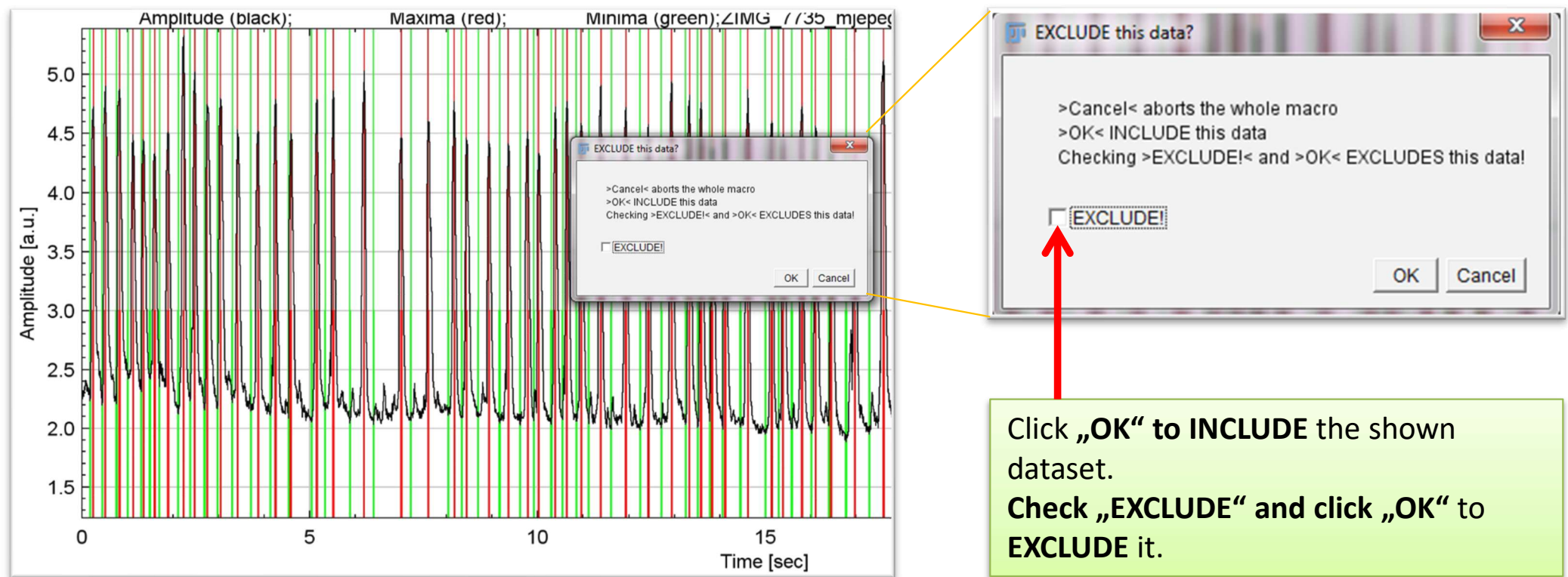
- Select „**5. Exclude Data**“ and click „**OK**“.
- **Just select** the „**dataplots**“ folder created after evaluation (**DO NOT open** it), containing the **amplitude plots** and the „**Results**“ (as text file).
- After selecting/excluding your data, you will be asked again to choose **another „dataplots“** folder to continue excluding until you terminate the macro.



The Workflow of MYOCYTER v2.0

5. Exclude Data

- Now, you will be asked to **decide for every single recognized cell** if the corresponding **data should be EXCLUDED**.



The Workflow of MYOCYTER v2.0

5. Exclude Data

- New files are generated in the „**dataplots**“ folder:
„**Results (selected)**“ – This file contains **only the data you included** as well as a **list of the excluded data**.
These **results are also saved** as **individual files** for **each cell**.
Furthermore, a file with **summary statistics** is created: „**Results (final summary)**“.
The **included amplitudes** are saved in a **large file** („**Amplitudes (selected)**“),
as well as in **separate files** for **every single amplitude**.
- You can now process/compare this data table in a stats software.
- The **original data remain unchanged**.

The Workflow of MYOCYTER v2.0

5. Exclude Data

- The **results table** can be **quite long** (several megabytes).

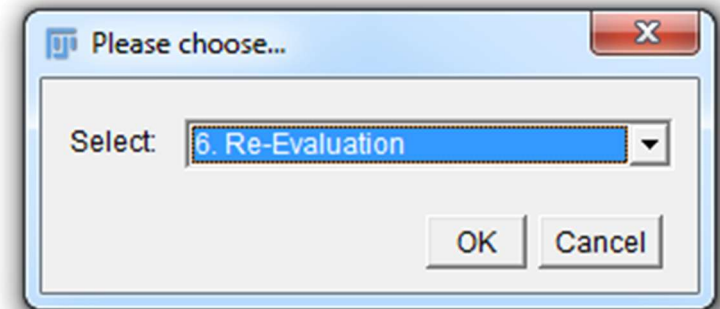
Please note: The **final statistics** are found **BOTH** under every single video evaluation (for every single cell) **AND at the very end of the table** also for every single video above, **AND as a separate file “Results (final summary)”**.

⇒ The user who only needs the statistics no longer has to laboriously gather the individual summary statistics from the whole table.

The Workflow of MYOCYTER v2.0

6. Re-Evaluation

- Using „**6. Re-Evaluation**“, you can apply changed settings to your data, **even after evaluation**.
- Select „**6. Re-Evaluation**“ and click „OK“.
- Go to the „**dataplots**“ subfolder of your evaluation, **open it**, select the „**Amplitudes only**“ or the „**Amplitudes only (selected)**“ text file, click „**OK**“ to open it.



The Workflow of MYOCYTER v2.0

6. Re-Evaluation

- Now, you are asked to **apply other settings** to your **evaluated data**.

Please apply your...

Detection	20
Threshold 1 [%]	10
Threshold 2 [%]	20
Threshold 3 [%]	50
Threshold 4 [%]	90
Framerate [fps]	120

OK Cancel

Higher values for „Detection“ include smaller local **maxima** as detected peaks/contractions.

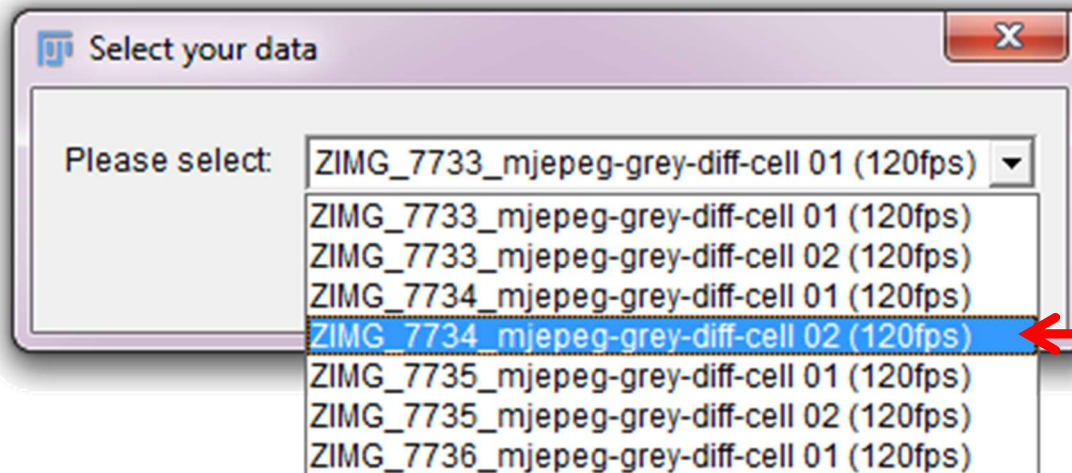
Apply **four different thresholds** to the extracted **amplitude**. Please see the manual for details.

Change the detected framerate of the video (if necessary).

The Workflow of MYOCYTER v2.0

6. Re-Evaluation

- Select the dataset (a single cell, in this case) you want to re-evaluate.

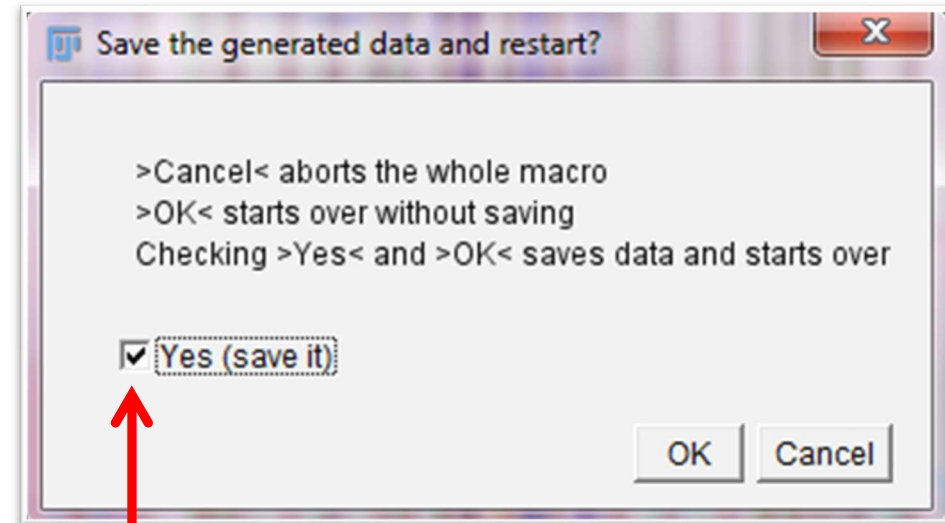
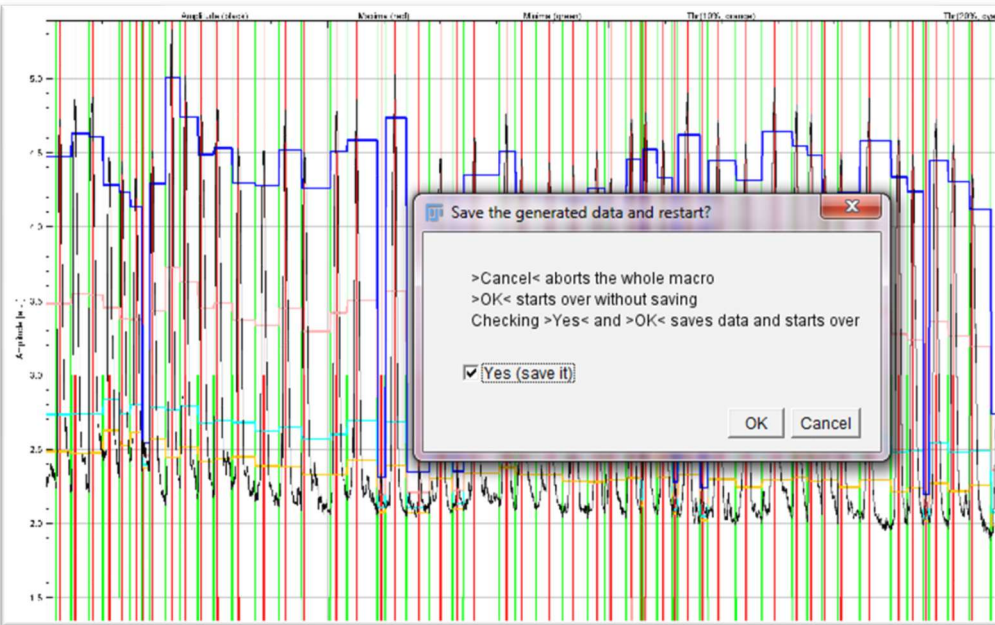


Every single cell from your evaluation is listed here – just pick one.

The Workflow of MYOCYTER v2.0

6. Re-Evaluation

- Save or discard the (new) **dataset after re-evaluation**:



Check „Yes“ and click „OK“ to save the new plot as well as the new results table.
Just click „OK“ to restart re-evaluation with other settings and data (without saving).

The Workflow of MYOCYTER v2.0

6. Re-Evaluation

- All **re-evaluated data** will be **stored** both **as plots**:
old filename + „re-evaluated“ and the current date and time (JPEG-format)

and also **as a single data table**:
„Amplitudes only“ + „re-evaluated“ and the current date and time (text file).

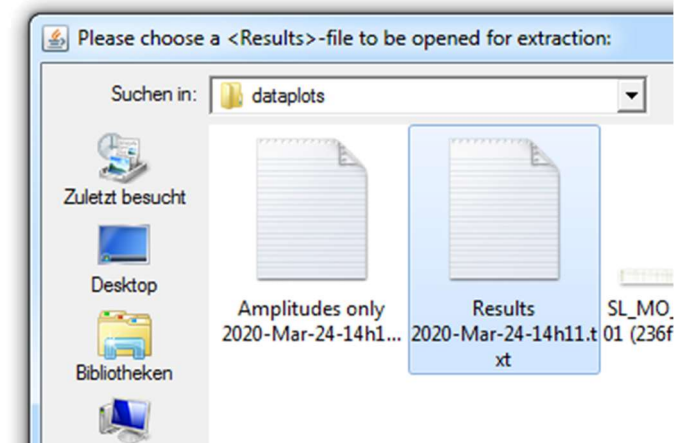
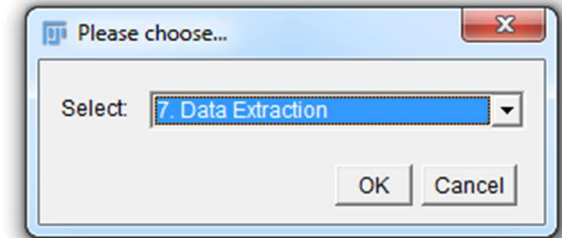
The Workflow of MYOCYTER v2.0

7. Data Extraction

Evaluation provides both „**continuous**“ data for plotting (like the amplitude over time) and „**discrete**“ data (like **only the maximum amplitudes** for every single contraction) for statistics.

This feature enables **extraction of ALL „discrete“ data from a „Results“-file** for further data processing in statistical software.

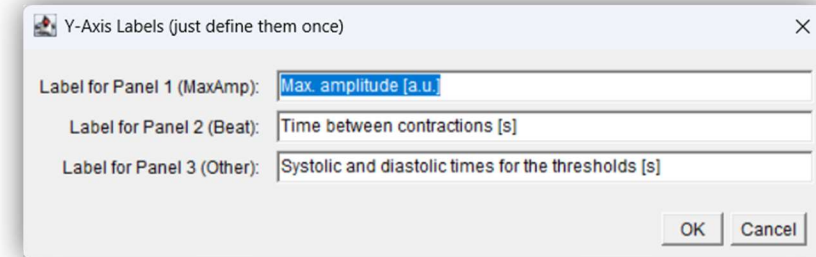
- First, you are asked to select the according „**Results**“- or „**Results (selected)**“-file (found in the „dataplots“-folder).
- Do **NOT** select the „Results (final summary)-file“.
- Klick „**OK**“ to open it.



The Workflow of MYOCYTER v2.0

7. Data Extraction

- You will be asked to **specify labels** for the **plots to be generated**; the suggested labels may be left unchanged. Examples of these plots are shown on the following pages.
- The **extracted data** („**Extracted results**“) are stored in the same folder as the according „**Results**“-file.
- **Only data** (i.e., **contractions**) are extracted that contain **all of the 14 parameters** (the **local maximum amplitude** of every recognized contraction, the **time since the last contraction**, as well as the **systole, diastole, and total time for each individual threshold**).
- After this, MYOCYTER will immediately ask for the **next „Results“- or „Results (selected)“-file to process** until you terminate the macro.





The Workflow of MYOCYTER v2.0

7. Data Extraction

Please note:

- **14 different discrete parameters** from the whole file will be extracted completely, **without any information about** the corresponding **files and recognized cells**.
- This **feature is only useful** if **ALL cells** in the according „Results“-file are **from the same sample** and should be compared to another sample (like „**Control**“ and „**Treatment**“).
- **Two different files** are generated and stored in the folder of the „Results“-file:
„**Extracted results (eng)**“ (**points as decimal separators**)
and
„**Extracted results (ger)**“ (**commas as decimal separators**)
- The data are arranged for **direct copy/paste transfer into a statistical software**.



Representative results table of the extracted parameters:

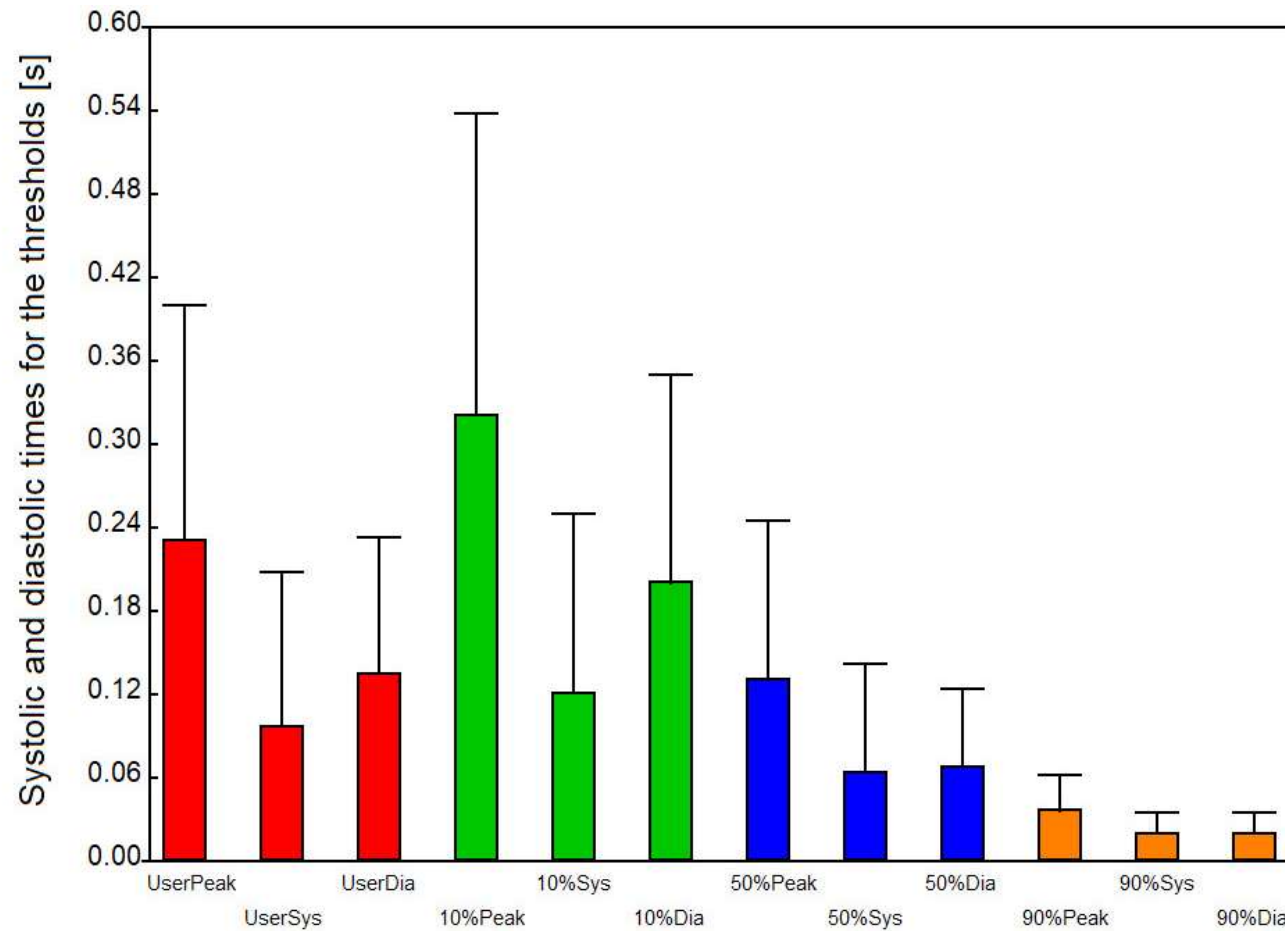
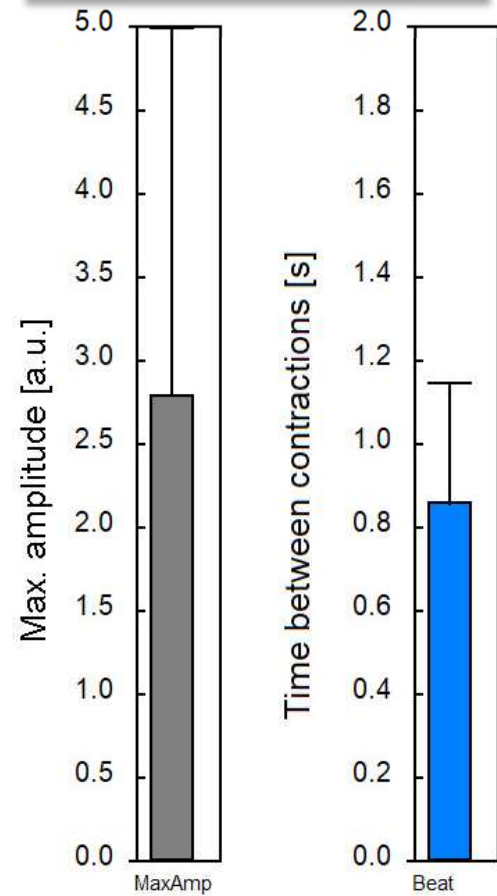
[illegible]



The Workflow of MYOCYTER v2.0

7. Data Extraction

For a quick overview, the extracted data is also displayed and saved as a plot.

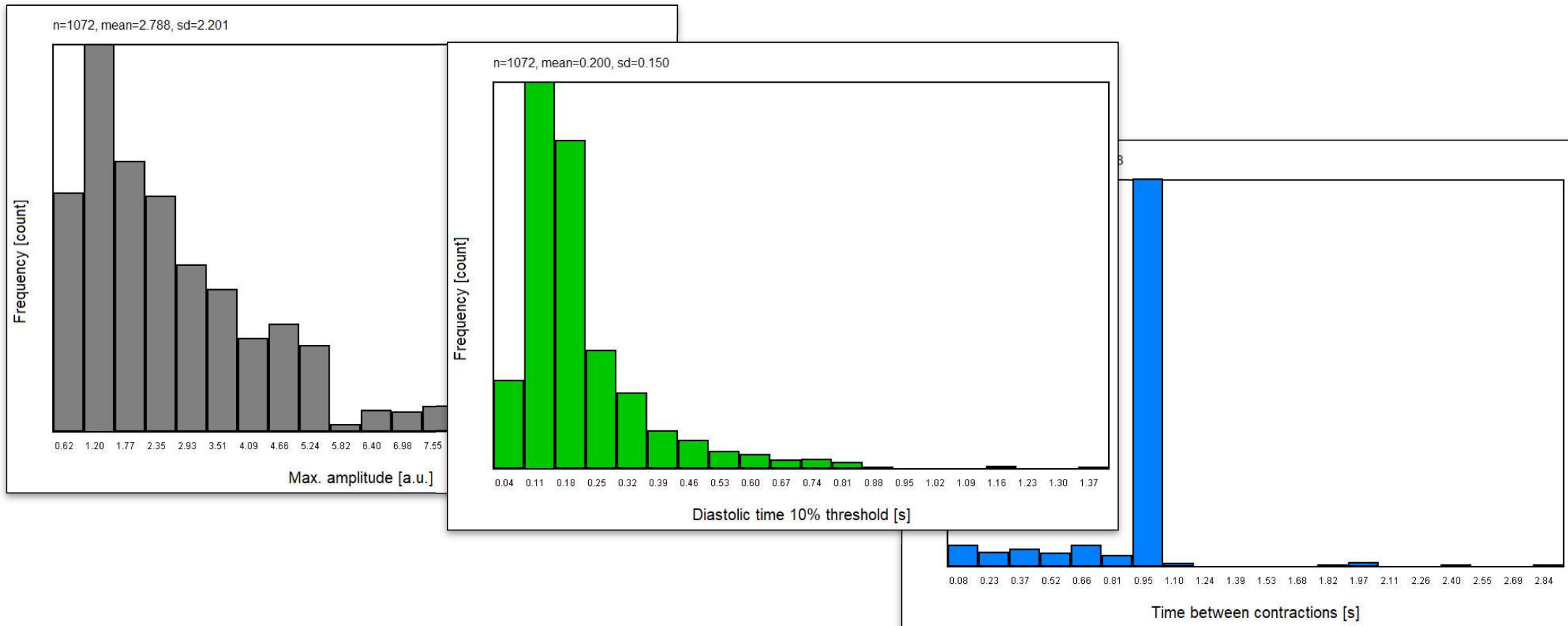




The Workflow of MYOCYTER v2.0

7. Data Extraction

Fourteen histograms showing the distributions of amplitudes, times between contractions, and peak times, systoles, and diastoles across all four thresholds are generated.

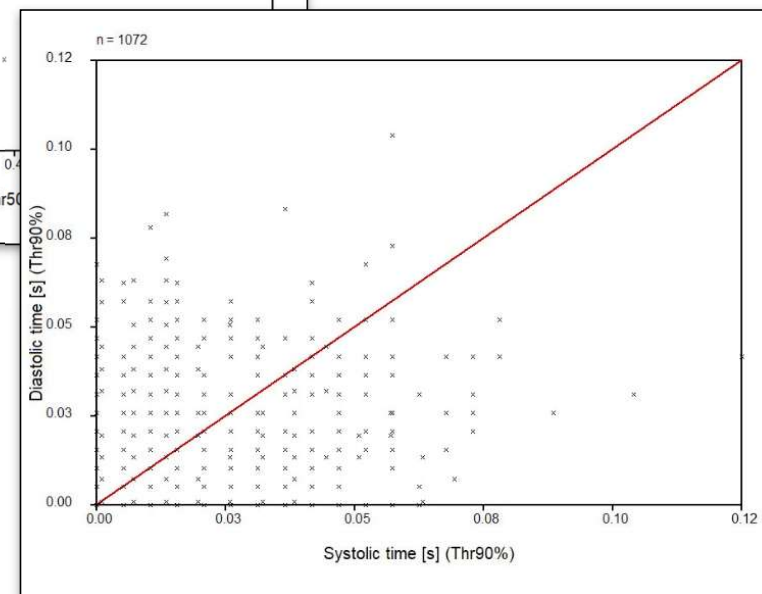
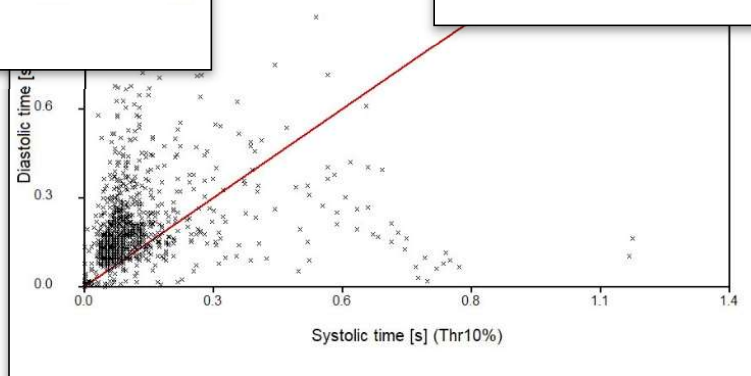
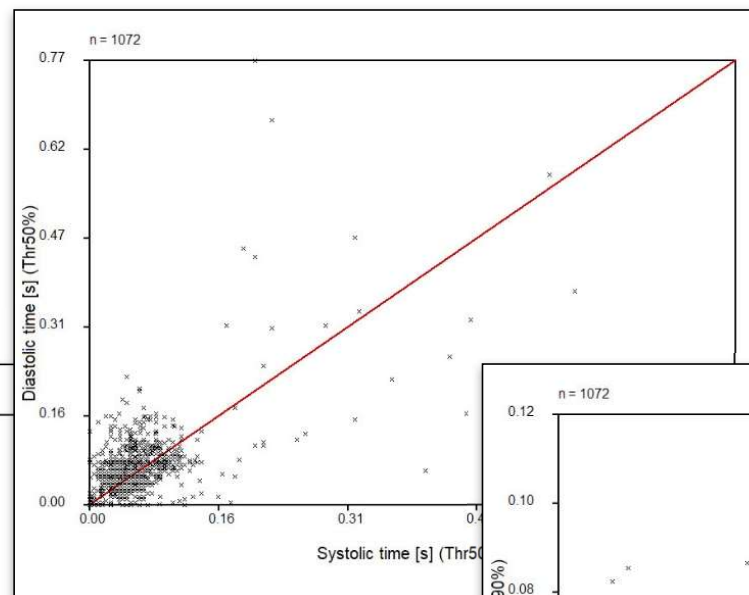
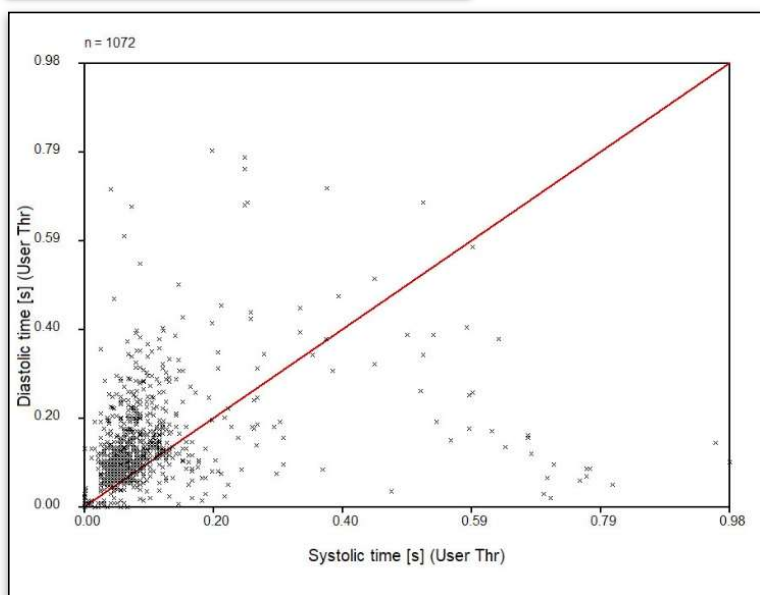




The Workflow of MYOCYTER v2.0

7. Data Extraction

Four scatterplots comparing systolic and diastolic times for all four thresholds.

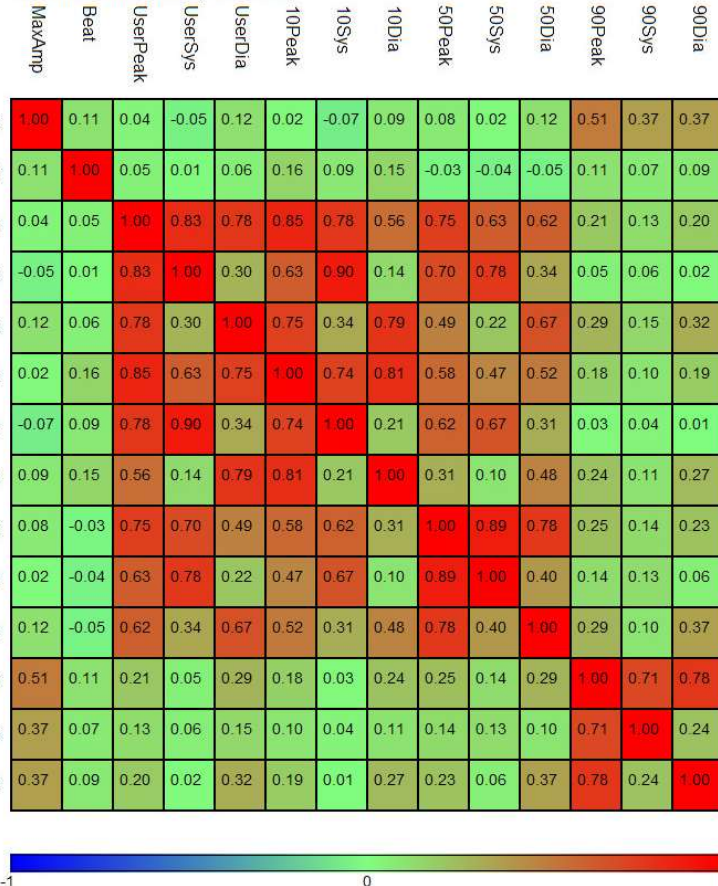




The Workflow of MYOCYTER v2.0

7. Data Extraction

Correlation heatmap (Pearson r)



A Correlation Heatmap (Pearson r)

This plot shows the pairwise **Pearson correlation coefficient** (r) between all measured parameters. Each cell represents the correlation between the parameter in the respective row (Y-axis) and the parameter in the respective column (X-axis).

x_i, y_i are paired data points;

$mean_x, mean_y$ are the

arithmetic means of x and y ;

Σ denotes the sum over all valid data pairs

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \cdot \sum_{i=1}^n (y_i - \bar{y})^2}}$$

Value range and meaning:

$r = 1.0 \rightarrow$ perfect **positive linear correlation** (red)

$r = 0.0 \rightarrow$ **no linear correlation** (green)

$r = -1.0 \rightarrow$ **perfect negative linear correlation** (blue)

How to read the heatmap:

Read across the row and column labels to find the parameter pair.

The number inside each cell is the **Pearson r value** for that pair.

Symmetry: the heatmap is mirrored along the diagonal; the diagonal cells ($r = 1$) are the **self-correlations of each parameter**.



Extraction for statistics

8. Extraction for Statistics

Select parameters to extract

Please select (only once) the parameters to be extracted

- ☒ Filenames
- ☒ Beats counted
- ☒ Amplitudes
- ☒ BeatTimes
- ☒ Frequencies
- ☒ Peaktime Thr10%
- ☒ Systole Thr10%
- ☒ Diastole Thr10%
- ☒ Peaktime user
- ☒ Systole user
- ☒ Diastole user
- ☒ Peaktime Thr50%
- ☒ Systole Thr50%
- ☒ Diastole Thr50%
- ☒ Peaktime Thr90%
- ☒ Systole Thr90%
- ☒ Diastole Thr90%

OK Cancel

At the first launch you'll be asked to pick the **parameters to be extracted** and the **dataplots** folder that holds the **"Results (final summary)"**-file (created in **5. Exclude data**). The tool writes an **"[Main Folder name]_Results Final Summary for Stats_[Date]"**-file, averaging the selected parameters for every cell across all videos in that folder.

Output is provided in **two different formats/files**:

English – decimal point

German – decimal comma



Extraction for statistics

8. Extraction for Statistics

You'll receive a table that lists **statistics for every parameter** you selected for extraction: **minimum** (Min), **maximum** (Max), **mean** and **standard deviation** (StdDev) of all contractions ("**Beats counted**") for each cell in each video (see "**Filenames**", listing both video and the corresponding cell). The columns are arranged as shown here:

Filenames	Beats counted [N]		Amplitudes [au]	Amplitudes [au]	Amplitudes [au]	Amplitudes [au]	All other → parameters [...]
			<i>Min</i>	<i>Max</i>	<i>Mean</i>	<i>StdDev</i>	[...]
IMG_3168_mjpeg-diff-cell 02 (170fps)	24		1,5825	3,1253	2,2994	0,4771	[...]
IMG_3169_mjpeg-diff-cell 01 (170fps)	26		0,5854	1,779	1,5979	0,2239	[...]
IMG_3169_mjpeg-diff-cell 03 (170fps)	18		2,6918	4,6689	3,4003	0,4399	[...]
IMG_3170_mjpeg-diff-cell 01 (170fps)	26		1,0875	4,4303	3,4095	0,7121	[...]
IMG_3172_mjpeg-diff-cell 01 (170fps)	32		0,7184	3,2031	1,7336	0,4973	[...]
IMG_3172_mjpeg-diff-cell 02 (170fps)	23		1,5566	2,8926	1,9526	0,2574	[...]
IMG_3173_mjpeg-diff-cell 01 (170fps)	23		2,7085	5,5051	3,4289	0,6529	[...]
IMG_3173_mjpeg-diff-cell 02 (170fps)	24		2,468	4,6428	3,3308	0,5729	[...]
IMG_3173_mjpeg-diff-cell 04 (170fps)	25		0,3029	1,1457	0,5828	0,1537	[...]

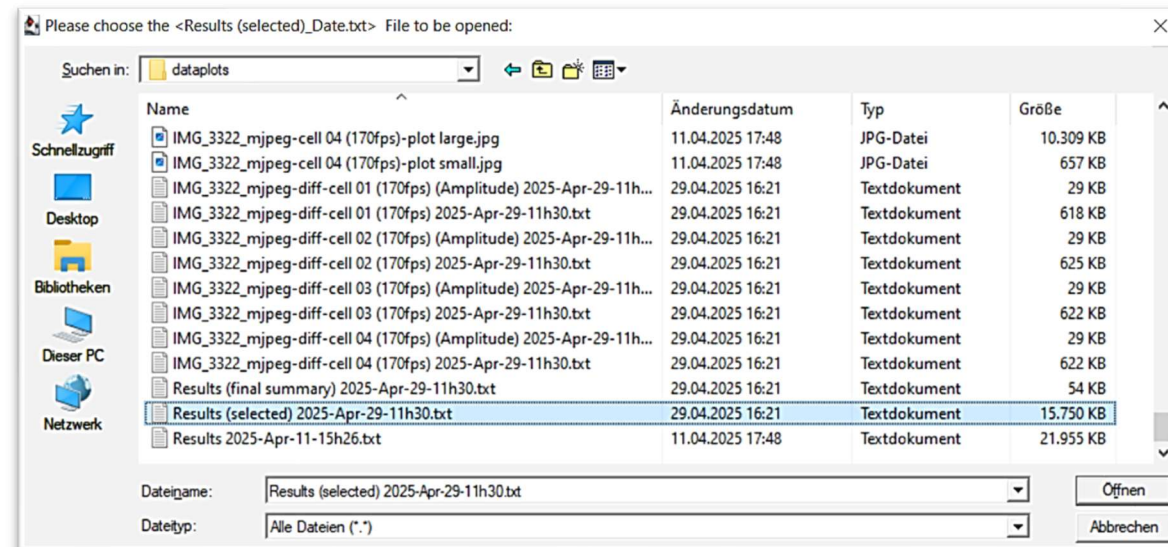


Statistics for selected samples

9. Statistics for selected Samples

This function **compares cells from the same Petri dish**, and the results are displayed as a heatmap. The considered parameters are the **amplitude**, the **systolic and diastolic durations** for all four thresholds, and the **mean value of all this data combined**.

First, select a “**Results (selected)_Date.txt**”-file from the corresponding “dataplots”-folder. The data will be processed, and a “**Statistics_Date_Time**”-folder containing the results (images and tables) will be **created within the “dataplots” folder**.





Statistics for selected samples

9. Statistics for selected Samples

Euclidean distance

$$D(i) = \sqrt{\sum_{k=1}^9 (P_k(i) - P_k(0))^2}$$

Relative Euclidean distance

$$D_{\text{rel}}(i) = \sqrt{\sum_{k=1}^9 \left(\frac{P_k(i)}{P_k(0)} \right)^2}$$

Mean absolute relative change

$$M(i) = \frac{1}{9} \sum_{k=1}^9 \left| \frac{P_k(i) - P_k(0)}{P_k(0)} \right|$$

Standard Deviation of Normalized Parameters

$$\sigma(i) = \sqrt{\frac{1}{9} \sum_{k=1}^9 \left(\frac{P_k(i)}{P_k(0)} - \bar{r}(i) \right)^2}, \quad \bar{r}(i) = \frac{1}{9} \sum_{k=1}^9 \frac{P_k(i)}{P_k(0)}$$

$$\sigma(i) = \sqrt{\frac{1}{9} \sum_{k=1}^9 (r_k(i) - \bar{r}(i))^2}$$



Statistics for selected samples

9. Statistics for selected Samples

Simplified Principal Component
(mean of absolute differences)

$$PC1(i) = \frac{1}{9} \sum_{k=1}^9 (P_k(i) - P_k(0))$$

Inter-beat Interval & Frequency

$$\Delta t_i = t_i - t_{i-1}, \quad f_i = \frac{1}{\Delta t_i}$$

Velocity & Acceleration

$$v_i = \Delta t_i, \quad a_i = v_i - v_{i-1}$$

Correlation Estimate (sum of products of selected contraction/relaxation differences)

$$C(i) = \sum_{m \in \{10\%, \text{User}, 50\%, 90\%\}} \left(P_{\text{Cont},m}(i) - P_{\text{Cont},m}(0) \right) \left(P_{\text{Rel},m}(i) - P_{\text{Rel},m}(0) \right)$$



Statistics for selected samples

9. Statistics for selected Samples

Cluster Assignment

$$\text{Cluster}(i) = \begin{cases} 1, & A(i) > 1.2 A(0) \wedge \Delta t_i > 1.2 \Delta t_1, \\ 2, & A(i) < 0.8 A(0) \wedge \Delta t_i < 0.8 \Delta t_1, \\ 3, & \text{otherwise.} \end{cases}$$

Moving Average over a window of size $2w+1$:

$$\text{MA}_i = \frac{1}{2w+1} \sum_{j=i-w}^{i+w} A(j)$$



ANOVA across cells from the same Petri

9. Statistics for selected Samples

Given G groups of sizes n_g with pooled data $x_{g,j}$:

1. Group means

$$\mu_g = \frac{1}{n_g} \sum_{j=1}^{n_g} x_{g,j},$$

$$\text{grand mean } \mu = \frac{1}{N} \sum_g \sum_j x_{g,j}, \quad N = \sum n_g$$

2. Sum of Squares

$$\text{Between: SSB} = \sum_{g=1}^G n_g (\mu_g - \mu)^2$$

$$\text{Within: SSW} = \sum_{g=1}^G \sum_{j=1}^{n_g} (x_{g,j} - \mu_g)^2$$

3. Degrees of freedom

$$df_B = G - 1, \quad df_W = N - G$$

4. Mean squares & F-statistic

$$\text{MSB} = \frac{\text{SSB}}{df_B}, \quad \text{MSW} = \frac{\text{SSW}}{df_W}, \quad F = \frac{\text{MSB}}{\text{MSW}}$$

5. Critical F at $\alpha = 0.05$: lookup F 0.05 (df_B, df_W)

If $F > F_{\text{crit}}$: significant



ANOVA – Post-Hoc Pairwise Tests

9. Statistics for selected Samples

Bonferroni-corrected t

For groups i, j :

$$t_{ij} = \frac{|\mu_i - \mu_j|}{\sqrt{\text{MSW} \left(\frac{1}{n_i} + \frac{1}{n_j} \right)}}, \quad t_{\text{crit}} = t_{df_W, 1-\alpha/(2 \binom{G}{2})}$$

Tukey-HSD q

$$q_{ij} = \frac{|\mu_i - \mu_j|}{\sqrt{\frac{1}{2} \text{MSW} \left(\frac{1}{n_i} + \frac{1}{n_j} \right)}}, \quad q_{\text{crit}} = q_{G, df_W, 0.05}$$

Results are visualized as heatmaps of t_{ij} or q_{ij} .

Composite Index of all 9 parameters

Normalize each parameter array X to

$$\hat{X}(i) = \frac{X(i) - \min X}{\max X - \min X}$$

Composite value

$$C(i) = \frac{1}{9} \sum_{k=1}^9 \hat{X}_k(i)$$

ANOVA on $C(i)$ follows the same steps as described above.



ANOVA – Post-Hoc Pairwise Tests

9. Statistics for selected Samples

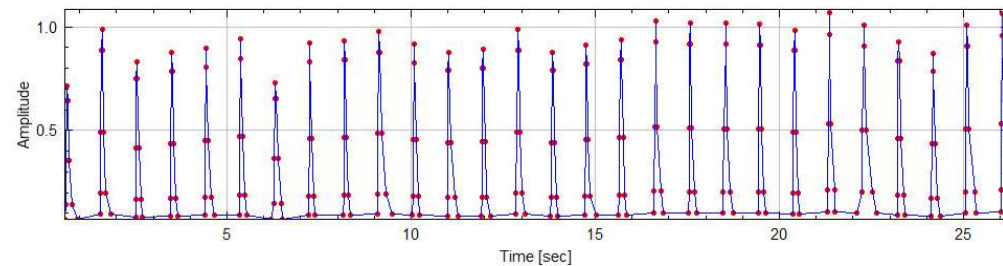
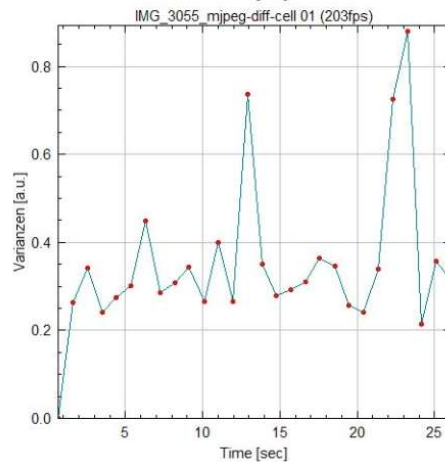
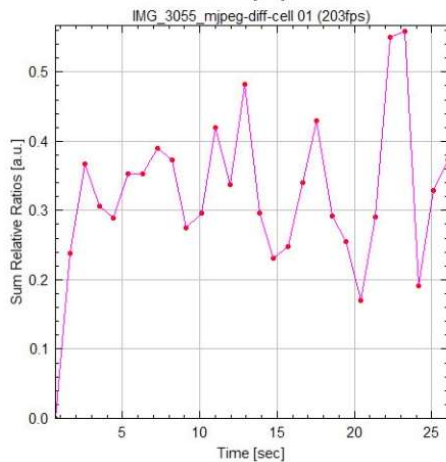
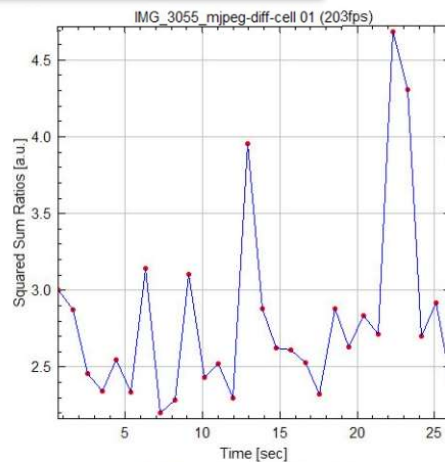
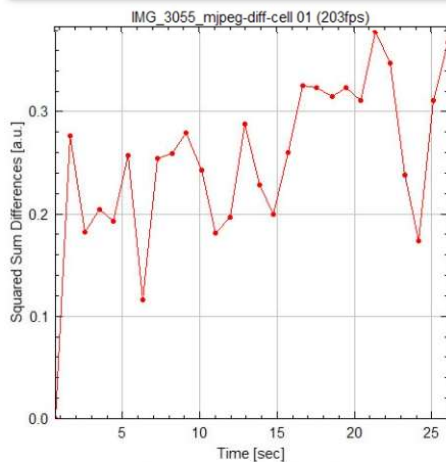
Squared Sum Differences: Euclidean distance of all 9 contraction parameters (absolute values) to the first contraction. Shows total absolute change; large deviations dominate due to squaring.

Squared Sum Ratios: Same as above but normalized to the first contraction (ratios). Shows relative (%-based) overall change.

Sum Relative Ratios: Mean absolute % deviation of all parameters from the first contraction; less sensitive to outliers.

Variances: Standard deviation of normalized parameters at each time point; measures spread between parameters' changes.

Amplitude: Time course of maximum contraction amplitude; direct measure of contraction strength.

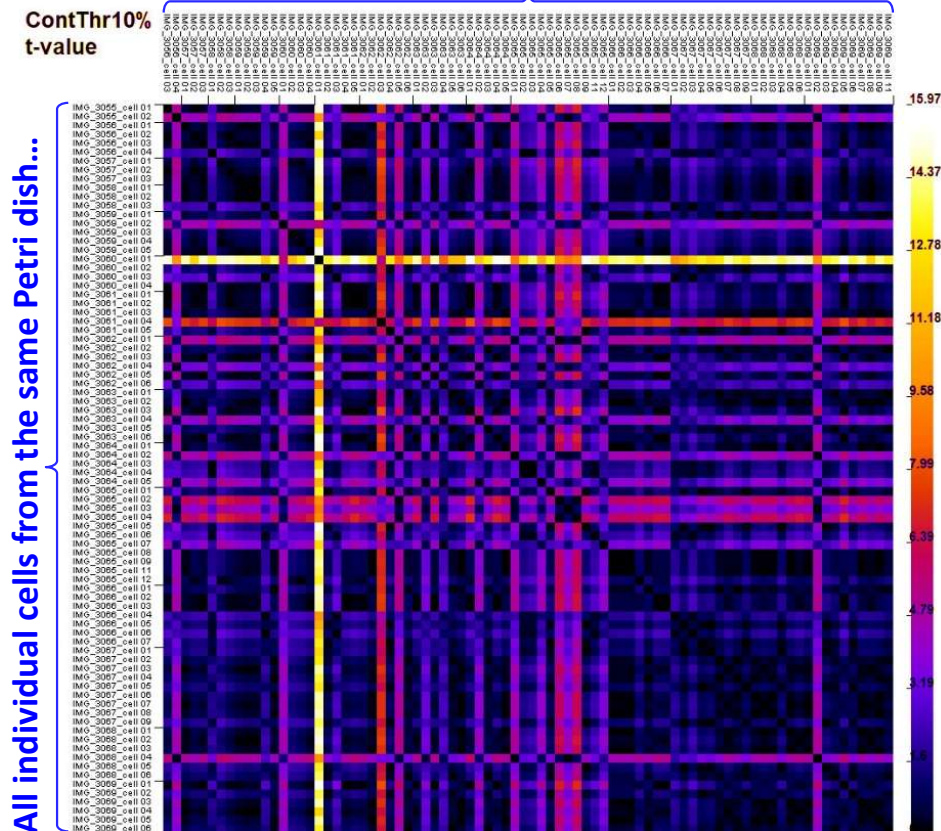




ANOVA – Post-Hoc Pairwise Tests

9. Statistics for selected Samples

...are compared with all other cells.



For each of the nine parameters (Amplitude, and Systole and Diastole at the 10%, user-defined, 50%, and 90% thresholds), a **separate heatmap** is generated, as well as a map **averaging all of the nine parameters**. Each heatmap displays all possible **pairwise comparisons between individual cells within one experiment or Petri dish**.

The color in each square represents the **t-value** (see previous slides) for the **corresponding cell pair**. The **black diagonal line** from the top left to the bottom right shows the **comparisons of each cell with itself** (these always show no statistical difference).

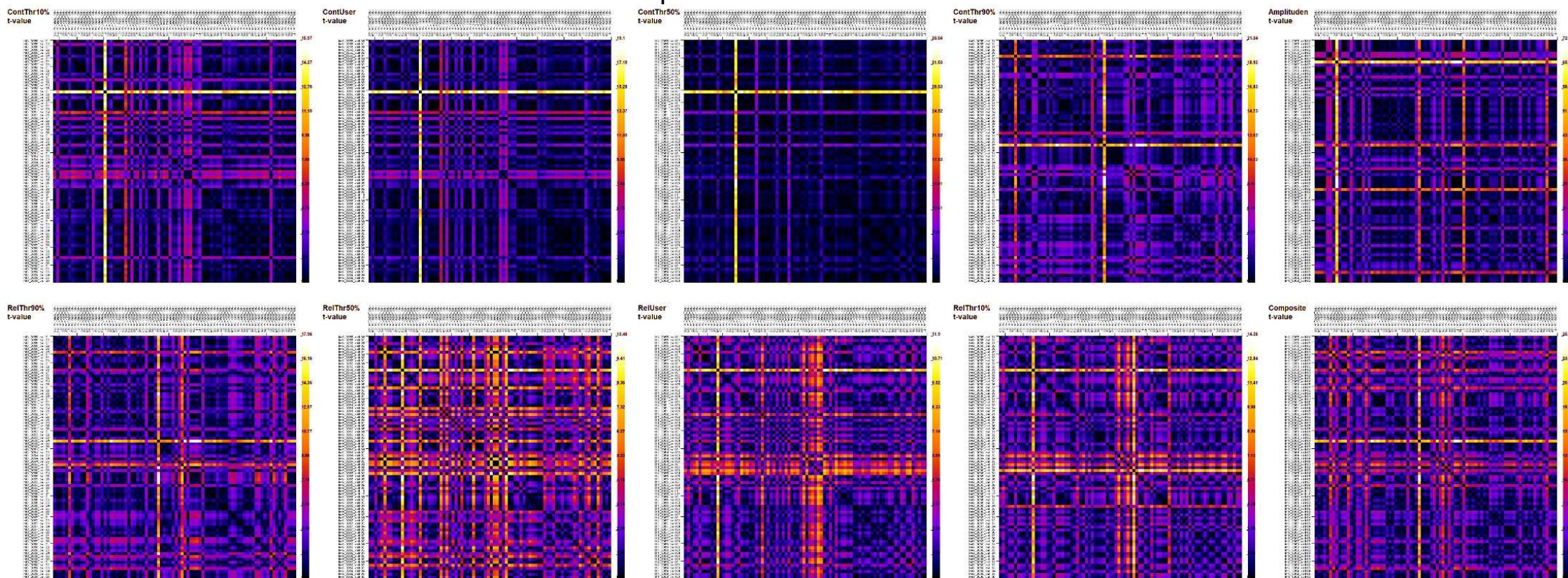
Each heatmap is **symmetrical along this diagonal**. This format allows for **easy identification of cells that differ significantly from all others**.



ANOVA – Post-Hoc Pairwise Tests

9. Statistics for selected Samples

Additionally, a summary figure is provided that displays **all ten individual heatmaps**, representing **each parameter** as well as the **overall mean** and allowing a comprehensive comparison of all cells within the same Petri dish or experiment.





ANOVA – Post-Hoc Pairwise Tests

9. Statistics for selected Samples

The “**Statistics_Date_Time**”-folder **created inside of the “dataplots”-folder** will also contain the **following new files**:

- Results ANOVA all Cells [Date-time] (ENG).txt
- Results ANOVA all Cells [Date-time] (GER).txt

- Results Final Summary for Stats [Date-time] (ENG).txt
- Results Final Summary for Stats [Date-time] (GER).txt

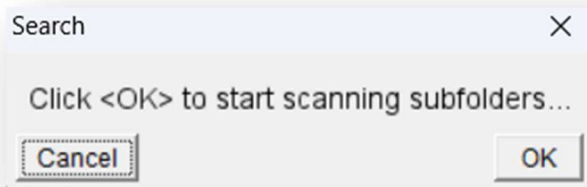


ANOVA across all samples

10. Overall Statistics (for cells)

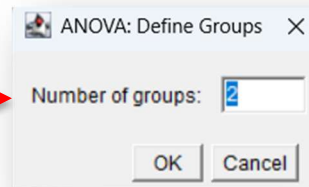
These are statistics for **CONTRACTIONS, AVERAGED OVER THE CORRESPONDING CELLS** as the smallest unit.

This function creates an **overall statistic** for all “Results Final Summary for Stats”-files located in a folder (select **ONLY** the main folder, **ALL** subfolders are automatically searched individually).



1. Start scanning the subfolders.

This will find every single “Results Final Summary for Stats”-file



2. Set the number of groups.

2 groups will be analyzed via Mann-Whitney-U-test,
 ≥ 3 via Kruskal-Wallis-test.

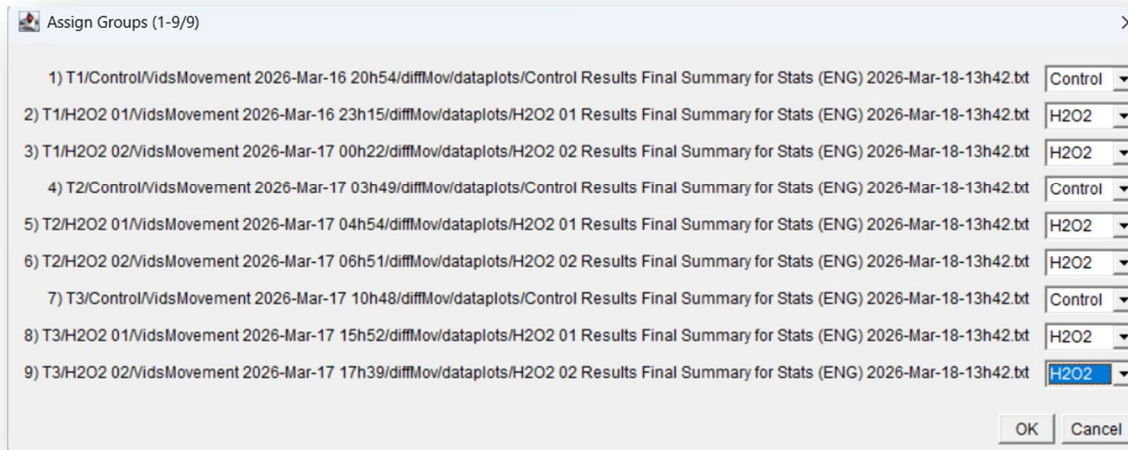


3. Define the group names.



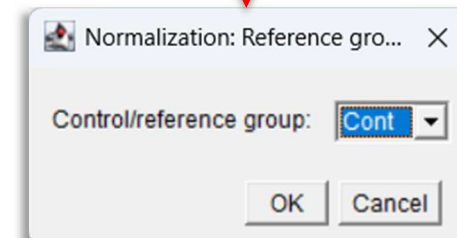
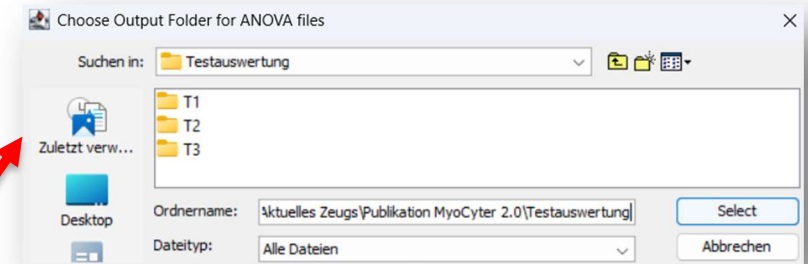
ANOVA across all samples

10. Overall Statistics (for cells)



4. Assign your samples to the corresponding groups.
You can also **exclude** single groups.

5. Select an output folder for the results.



6. Select the reference group.

Usually the control group, which will be used to normalize the data in the same folder (i.e. animal, see page 4). This normalization may be required to compare different animals, samples, or experiments.



In the selected output folder, **the following new files will be created:**

- **ANOVA_Ready_Data_[Date] LONG (ENG).txt** (all cell values are *listed in sequence*, for further processing in MyoCyter).
- **ANOVA_Ready_Data_[Date] LONG (GER).txt**
- **ANOVA_Ready_Data_[Date]_WIDE_ALLPARAMS (ENG).txt** (all cell values are *listed in adjacent columns* for export to third-party software).
- **ANOVA_Ready_Data_[Date]_WIDE_ALLPARAMS (GER).txt**

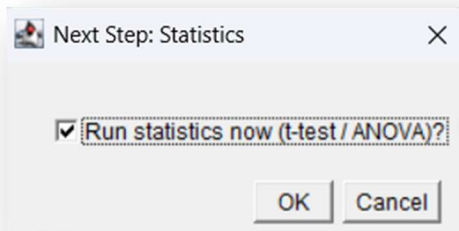
These files contain statistics derived from all complete contractions and averaged across the respective individual cells:

[illegible]

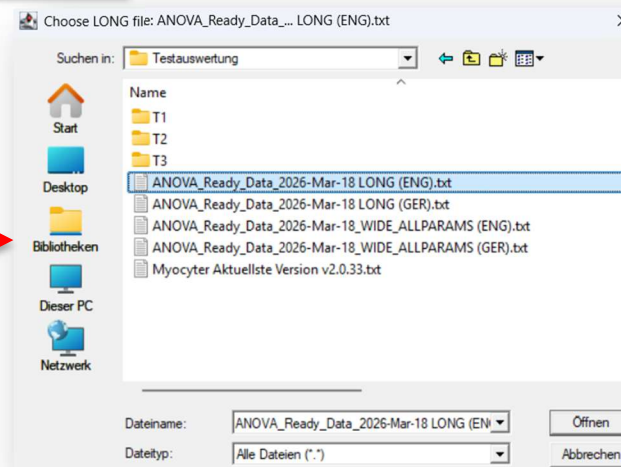


ANOVA across all samples

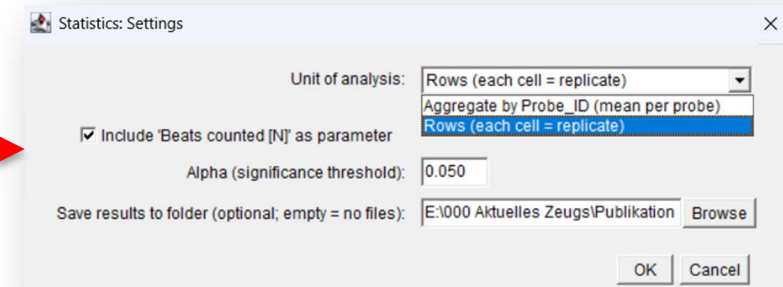
10. Overall Statistics (for cells)



6. Proceed with statistics.



7. Select the proper „ANOVA_Ready_Data_LONG (ENG).txt“-file that has just been created.



8. Select the proper statistical unit.

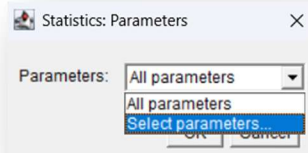
Rows (each cell = replicate): each cell is treated as a separate measurement.

Aggregate by Probe_ID: all cells in a sample are first averaged for each parameter; only these averages are included in the statistics. $p \leq \text{Alpha}$ will be considered as statistically significant.

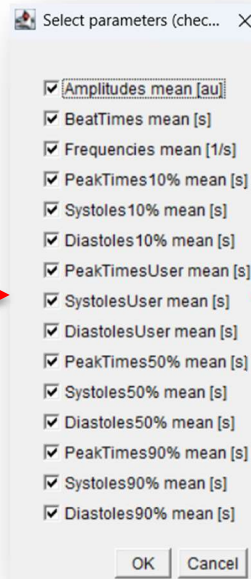


ANOVA across all samples

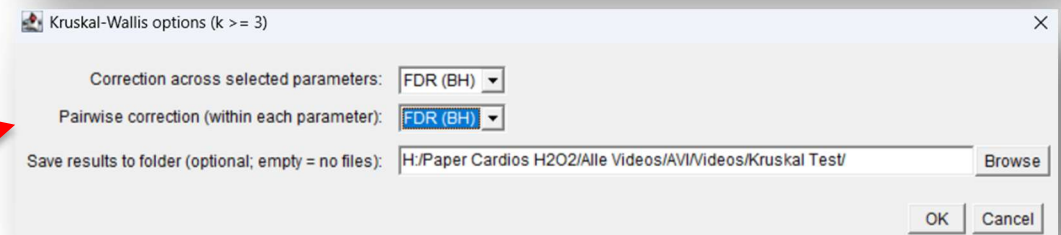
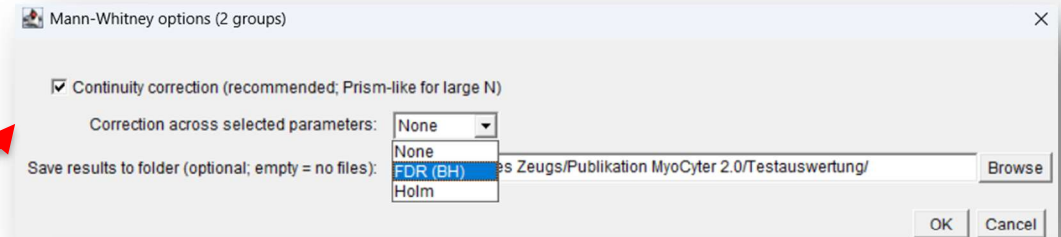
10. Overall Statistics (for cells)



9. Use all or select single parameters.



10. Include single parameters (optional).



11. Select the type of correction:

None: Higher sensitivity, but a greater risk of false-positive findings.

FDR (BH): Benjamini-Hochberg correction for controlling the false discovery rate; suitable when analyzing multiple parameters and a balanced trade-off between discovery rate and error control is desired.

Holm: Stricter control of erroneous decisions; more conservative than FDR, resulting in a lower risk of false-positive results.

For ≥ 3 groups: Across selected parameters: FDR (BH) → okay
Pairwise correction within each parameter: Holm or FDR (BH) → both okay.



ANOVA across all samples

10. Overall Statistics (for cells)

For 2 groups, the files

“STATS_mannwhitney_ANOVA_Ready_Data_[Date] LONG (ENG).txt” and
“STATS_mannwhitney_ANOVA_Ready_Data_[Date] LONG (GER).txt”
are generated.

For ≥ 3 groups, the files

“STATS_kruskalwallis__ANOVA_Ready_Data_2026-Mar-19 LONG (ENG).txt”
“STATS_kruskalwallis__ANOVA_Ready_Data_2026-Mar-19 LONG (GER).txt”
are generated.

The exact mathematics behind the tests is explained on the following pages.



ANOVA across all samples

10. Overall Statistics (for cells)

For each sample s and parameter p :

1. Count

$$N_s = |\{i : x_{p,i}^{(s)} \text{ defined}\}|$$

2. Mean

$$\bar{x}_p^{(s)} = \frac{1}{N_s} \sum_{i=1}^{N_s} x_{p,i}^{(s)}$$

3. Standard deviation

$$\sigma_p^{(s)} = \sqrt{\frac{1}{N_s} \sum_{i=1}^{N_s} (x_{p,i}^{(s)} - \bar{x}_p^{(s)})^2}$$

4. Append $\bar{x}_p^{(s)}$ and $\sigma_p^{(s)}$ to result arrays, and track $\text{GlobalMaxMean} = \max_{s,p} \bar{x}_p^{(s)}$.

5. **Concatenate all values** for parameter p across every sample into one big array of length

$$N_{\text{all}} = \sum_s N_s, \text{ then calculate}$$

Global mean $\mu_p = \frac{1}{N_{\text{all}}} \sum_{s=1}^S \sum_{i=1}^{N_s} x_{p,i}^{(s)}$

Global standard deviation $\sigma_p = \sqrt{\frac{1}{N_{\text{all}}} \sum_{s=1}^S \sum_{i=1}^{N_s} (x_{p,i}^{(s)} - \mu_p)^2}$



ANOVA across all samples

10. Overall Statistics (for cells)

Heatmap of sample means

Scaling factor

$$M = \left\lfloor \frac{250}{\max_{s,p} \bar{x}_p^{(s)}} \right\rfloor, \quad M \geq 1$$

Pixel intensity for sample s , parameter p

$$I_{s,p} = M \bar{x}_p^{(s)}$$

Visualize as $P \times S$ image (parameters $\times$ samples), apply „Fire“ LUT

Difference Map to Global Mean

Absolute deviation

$$d_{s,p} = |\bar{x}_p^{(s)} - \mu_p|$$

Pixel intensity

$$I_{s,p}^{\text{diff}} = M d_{s,p} = M \sqrt{(\bar{x}_p^{(s)} - \mu_p)^2}$$



ANOVA across all samples

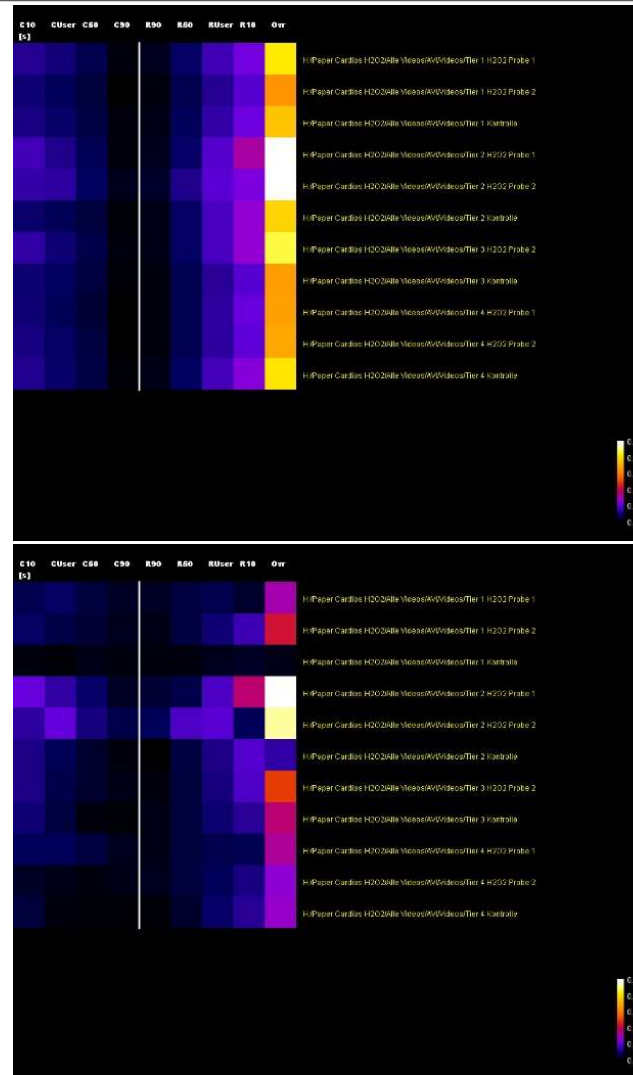
10. Overall Statistics (for cells)

The final statistics are presented as **two heat-maps**:

All samples (top) – mean duration (in seconds) for each parameter:
C10 to C90 (contraction, 10 to 90 %) and *R90 to R10* (relaxation, 90 to 10 %).

Sample vs. overall mean (bottom) – deviation of each sample's mean from the global mean (of all animals within the group).

Here, “*sample*” is all cells recorded in the same Petri dish/folder.



HeatMap all Samples.jpg

Animal 1, Sample 1 (Foldername and path)
Animal 1, Sample 2 (Foldername and path)
Animal 1, Sample 3 (Foldername and path)
...
Animal 2, Sample 1 (Foldername and path)
Animal 2, Sample 2 (Foldername and path)
Animal 2, Sample 3 (Foldername and path)
...

Differences to OverallMean.jpg

Animal 1 and Sample 1 (Foldername and -path)
Animal 1 and Sample 2 (Foldername and -path)
Animal 1 and Sample 3 (Foldername and -path)
...
Animal 2 and Sample 1 (Foldername and -path)
Animal 2 and Sample 2 (Foldername and -path)
Animal 2 and Sample 3 (Foldername and -path)
...



Two accompanying data tables are generated – “Final Overall Analysis [Date] (ENG/GER)”. They report mean and standard deviation across all cells within each sample/folder for direct comparison, including the overall contraction time (Ovr) in seconds. The tables include the underlying values for “HeatMap all Samples.jpg” as well as the values and absolute deviations (vs. the global mean) used for “Differences to OverallMean.jpg”.

[illegible]



ANOVA across all samples

10. Overall Statistics (for cells)

Normalization:

After group assignment, MYOCYTER can normalize cell-level mean values to a user-defined reference group, usually the control group. Normalization is performed separately within each experimental parent folder, so samples from different animals or independent experiments are not mixed.

For each parameter, the median of the reference/control cells within the same parent folder is calculated and used as the normalization reference. Beats counted [N] is retained for orientation but is not normalized.

Important: the current normalization module does not write normalized LONG tables. It writes block-wide normalized tables, with the reference/control group on the left and the treated/comparison group(s) on the right.

Non-normalized ANOVA-ready output files are still generated:

ANOVA_Ready_Data_[Date] LONG (ENG/GER).txt

ANOVA_Ready_Data_[Date]_WIDE_ALLPARAMS (ENG/GER).txt

ENG uses decimal points; GER uses decimal commas.



ANOVA across all samples

10. Overall Statistics (for cells)

Normalization output files:

Normalized_RATIO_raw_div_ControlMedian_BLOCKWIDE_ControlLeft_TreatedRight_[Date] (ENG/GER).txt

Contains the ratio-normalized data in block-wide format. Each row aligns cells within the same experimental parent folder. Parameter values are divided by the corresponding control median from that parent folder. A value of 1 therefore represents the control median; values >1 are above and values <1 are below the control median.

The table includes Experiment_Parent_Folder and Row_in_Parent, followed by group-specific columns such as [Group]__Probe_ID, [Group]__Filename, Beats counted [N]__[Group], and the normalized parameter columns.

Normalized_LOG2_raw_div_ControlMedian_BLOCKWIDE_ControlLeft_TreatedRight_[Date] (ENG/GER).txt

Contains the same block-wide data after log2 transformation: $\log_2(\text{raw value} / \text{control median})$. In this format, the control median is 0; positive values indicate values above the control median and negative values indicate values below it.



ANOVA across all samples

10. Overall Statistics (for cells)

Normalization reference and warning files:

Normalization_REFERENCES_ControlMedian_[Date] (ENG/GER).txt

Lists the medians used for normalization. Columns: Parent_Folder, Control_Group, Parameter, N_Control_Cells, Control_Median, Status. Use this file to verify which control medians were applied in each parent folder.

Normalization_WARNINGS_ControlMedian_[Date] (ENG/GER).txt

Lists potential problems detected during normalization, for example non-numeric values, NaN/Infinity entries, missing or invalid control medians, and rows excluded from normalization. The file also reports summary lines such as rows processed and parent folders detected.

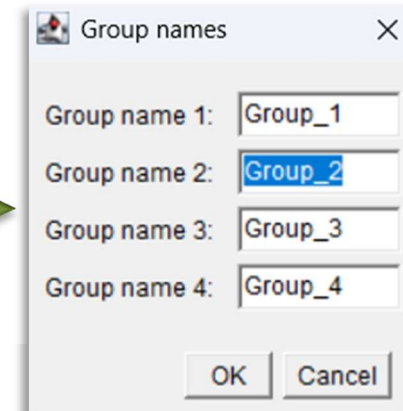
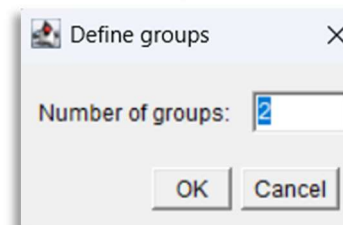
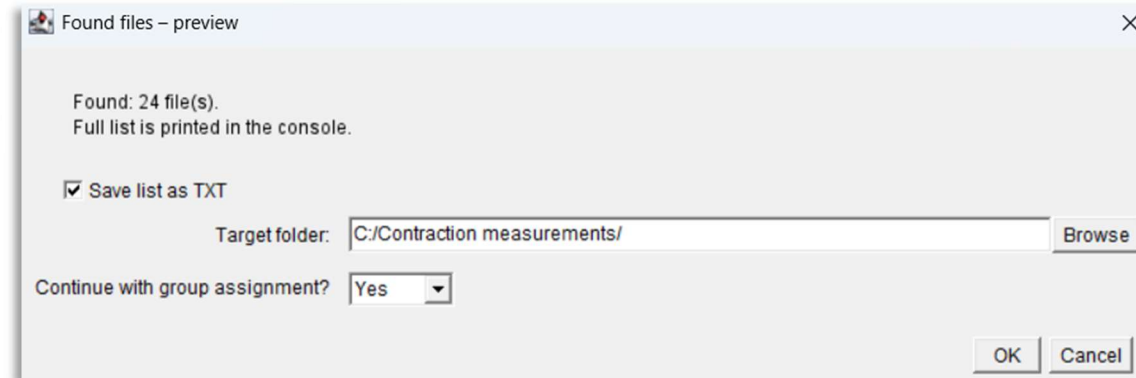
Note for release checking: if the filename contains _NaN instead of a date stamp, the date label was not resolved correctly during export. This affects the filename only and should be corrected in the macro before release; it is not the intended manual filename pattern.



ANOVA across all samples

11. Prepare data for ANOVA (contractions)

- Choose a **main/root directory** (just select, do **NOT** open).
 - The macro scans recursively (depth-first, iteratively) across **ALL subfolders**.
 - In each folder, it keeps **only the most recent file whose filename contains Extracted results (eng)** (case-sensitive).
 - The full list is printed in the Log, including each file's relative folder (relative to the chosen root).
 - Empty/inaccessible folders are noted in the Log and skipped.
- Define the **number of groups** (four, in this example).
- Name the according groups.





ANOVA across all samples

11. Prepare data for ANOVA (contractions)

- **Assign files to groups**
 - Files appear in chunks of 20 dialogs.
 - For each file choose: **one of your group names**, or **“Exclude”** (to drop the file).
- **Choose output folder**
 - Dialog: “Choose output folder (empty = use main folder)”.
 - Paths are normalized; trailing slash is enforced.
- **Written output**
 - For each group: **“Group_[Name]_merged (ENG/GER).txt”**
 - For all groups: **“ALL_groups_MERGED (ENG/GER).txt”**
 - A list of all **“FOUND_files.txt”**

Assignment (1–20 / 24)

1) 62/AVI/VidsMovement 2025-Apr-14 16h26/diffMov/dataplots/Extracted results (eng) Results (selected) 2025-May-05-14h32.bt	Male Cont
2) 63/AVI/VidsMovement 2025-Apr-23 16h05/diffMov/dataplots/Extracted results (eng) Results (selected) 2025-May-05-14h32.bt	Male Cont
3) 66/AVI/VidsMovement 2025-Apr-25 17h20/diffMov/dataplots/Extracted results (eng) Results (selected) 2025-May-05-14h32.bt	Male Cont
4) 67/AVI/VidsMovement 2025-Apr-22 09h58/diffMov/dataplots/Extracted results (eng) Results (selected) 2025-May-05-14h32.bt	Female Noise
5) 70/AVI/VidsMovement 2025-Apr-25 12h18/diffMov/dataplots/Extracted results (eng) Results (selected) 2025-May-05-14h32.bt	Male Cont
6) 72/AVI/VidsMovement 2025-Apr-17 18h10/diffMov/dataplots/Extracted results (eng) Results (selected) 2025-May-05-14h32.bt	Female Noise
7) 74/AVI/VidsMovement 2025-Apr-17 09h17/diffMov/dataplots/Extracted results (eng) Results (selected) 2025-May-05-14h32.bt	Male Noise
8) 76/AVI/VidsMovement 2025-Apr-15 10h56/diffMov/dataplots/Extracted results (eng) Results (selected) 2025-May-05-14h32.bt	Male Cont
9) 30/AVI/VidsMovement 2025-Apr-28 17h15/diffMov/dataplots/Extracted results (eng) Results (selected) 2025-Aug-21-11h51.bt	Male Cont
10) 31/AVI/VidsMovement 2025-Mar-25 08h45/diffMov/dataplots/Extracted results (eng) Results (selected) 2025-Aug-20-15h11.bt	Female Noise
11) 34/AVI/VidsMovement 2025-Mar-27 16h27/diffMov/dataplots/Extracted results (eng) Results (selected) 2025-Apr-28-17h09.bt	Male Cont
12) 35/AVI/VidsMovement 2025-Apr-03 11h06/diffMov/dataplots/Extracted results (eng) Results (selected) 2025-Apr-29-11h30.bt	Male Cont
13) 38/AVI/VidsMovement 2025-Apr-29 11h54/diffMov/dataplots/Extracted results (eng) Results (selected) 2025-May-06-16h59.bt	Female Cont
14) 39/AVI/VidsMovement 2025-Apr-03 20h11/diffMov/dataplots/Extracted results (eng) Results (selected) 2025-Apr-29-11h30.bt	Male Cont
15) 42/AVI/VidsMovement 2025-Apr-07 09h59/diffMov/dataplots/Extracted results (eng) Results (selected) 2025-Apr-29-11h30.bt	Male Cont
16) 43/AVI/VidsMovement 2025-Apr-10 16h07/diffMov/dataplots/Extracted results (eng) Results (selected) 2025-Apr-29-11h30.bt	Exclude
17) 46/AVI/VidsMovement 2025-Apr-10 11h52/diffMov/dataplots/Extracted results (eng) Results (selected) 2025-Apr-29-11h30.bt	Male Cont
18) 48/AVI/VidsMovement 2025-Apr-11 12h03/diffMov/dataplots/Extracted results (eng) Results (selected) 2025-Apr-29-11h30.bt	Female Cont
19) 50/AVI/VidsMovement 2025-Apr-11 15h26/diffMov/dataplots/Extracted results (eng) Results (selected) 2025-Apr-29-11h30.bt	Male Cont
20) 52/AVI/VidsMovement 2025-Apr-11 19h05/diffMov/dataplots/Extracted results (eng) Results (selected) 2025-Apr-29-11h30.bt	Male Cont

OK Cancel

“ALL_groups_MERGED (ENG).txt” is essential for the next step: “12. Apply ANOVA”.



ANOVA across all samples

12. Apply ANOVA (contractions)

These are statistics for SINGLE CONTRACTIONS as smallest unit.

1. Input and parsing:

Choose the text file named **“ALL_groups_MERGED (ENG).txt”**, created via **“11. Prepare data for ANOVA (contractions)”**.

This file contains 14 numeric parameters to be compared. The exact header names are preserved and used in the output. The macro requires ≥ 2 groups overall; otherwise it aborts.

2. Mode selection (automatic):

Two groups detected ($k = 2$): runs Welch's t-test (two-sided) for each parameter; this is robust to unequal variances and unequal sample sizes.

Three or more groups ($k \geq 3$): runs a one-way ANOVA per parameter and (optionally) pairwise Welch tests within each parameter.

You don't choose the mode, the code chooses based on how many unique group labels it finds.



ANOVA across all samples

12. Apply ANOVA (contractions)

3. Options dialog:

When you launch the analysis, you'll see the **Statistics Options dialog**.

Omnibus correction (across the 14 parameters): Applies a multiple-testing correction across parameters (i.e., you have 14 tests, one per parameter). Choices:

None: no cross-parameter correction.

FDR (BH): Benjamini-Hochberg **False Discovery Rate** control.

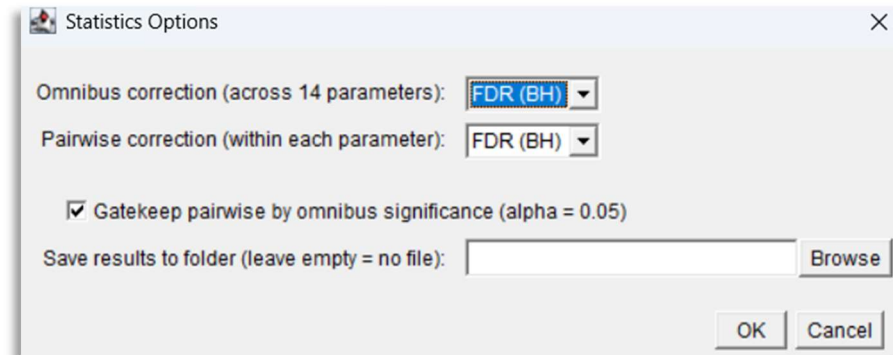
Use when your goal is to limit the expected proportion of false positives among all “significant” parameters; more liberal/powerful than Holm, suitable for exploratory screens where many parameters may show effects.

Holm: Holm's step-down **FWER (Family-Wise Error Rate)** control.

Use when you need strict family-wise error control across the 14 parameters (confirmatory analyses). It's more conservative than FDR, so you'll get fewer positives but with stronger error control.

In **two-group mode**, this correction is applied to the 14 t-test p-values (one per parameter).

In **≥3-group mode**, it's applied to the 14 ANOVA p-values (the “omnibus” tests).





ANOVA across all samples

12. Apply ANOVA (contractions)

Pairwise correction (within each parameter):

Only relevant when $k \geq 3$ (**ANOVA mode**). After the omnibus test for a given parameter, you may test all pairwise group contrasts within that parameter:

None: raw p-values for all pairwise Welch t-tests (many comparisons $\Rightarrow$ increased false positives).

FDR (BH): controls FDR within each parameter's set of pairwise tests.

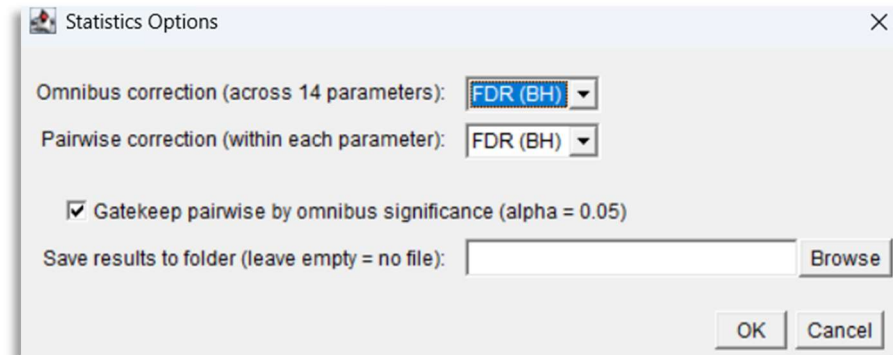
Holm: controls FWER within each parameter's pairwise tests using a **step-down Bonferroni procedure** (sequentially rejective), which is more powerful than simple Bonferroni.

Choose FDR if you're exploring many group differences; **choose Holm** if you want strong per-parameter family-wise error control.

Gatekeep pairwise by omnibus significance ($\alpha = 0.05$): **Only relevant when $k \geq 3$.**

If ON, the macro performs pairwise comparisons only if the omnibus ANOVA for that parameter is significant at $\alpha = 0.05$. This reduces multiplicity and guards against chasing noise when the overall difference isn't supported.

The **gatekeeping check** uses corrected omnibus p-values if you chose an omnibus correction; otherwise it uses raw omnibus p-values.





ANOVA across all samples

12. Apply ANOVA (contractions)

Save results to folder:

If you **provide a directory**, the macro writes **two files**:

- **STATS_ttest (ENG/GER).txt** (for 2 groups).
- **STATS_anova (ENG/GER).txt** (for ≥ 3 groups).

Leave empty to suppress file export (results still print in ImageJ's Log).

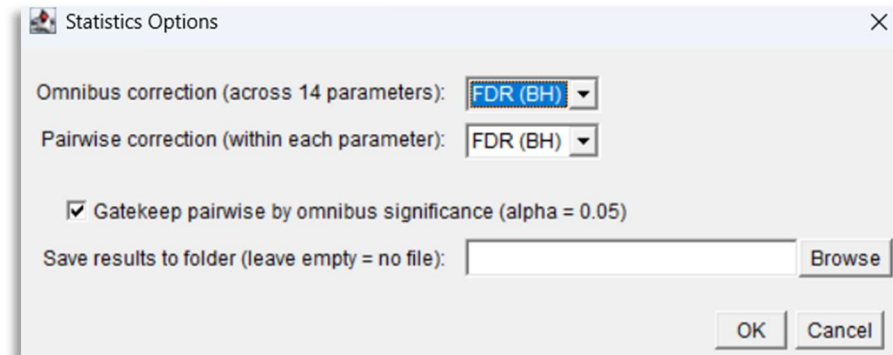
What the module computes:

Two groups: Welch's t-tests

Welch t-statistic, with **Welch-Satterthwaite** degrees of freedom (non-integer), robust to unequal variances.

- **Two-sided p-value** from the t distribution.
- **Omnibus multiple-testing correction** across the 14 parameter-level p-values:
- **FDR (BH)**: controls the expected false discovery rate across parameters.
- **Holm**: controls family-wise error across parameters.

$$t = \frac{\bar{x}_2 - \bar{x}_1}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}$$





ANOVA across all samples

12. Apply ANOVA (contractions)

≥3 groups: (One-way ANOVA + optional pairwise)

For each parameter:

- **One-way ANOVA** (fixed effects):
 - Sum of squares between groups (**SSB**), within groups (**SSW**), total (**SST**).
 - $F = MS_B / MS_W$ with $df_1 = k-1$ and $df_2 = N-k$.
 - **p-value** from the F-distribution.
 - Effect size: $\eta^2 = \frac{SS_B}{SS_T}$ (proportion of variance explained by group).
- **Omnibus multiple-testing correction** across the 14 ANOVA p-values (as above).
- **Gatekeeping (optional)**: if the omnibus is not significant (at $\alpha = 0.05$), pairwise tests are skipped for that parameter.



ANOVA across all samples

12. Apply ANOVA (contractions)

- **Pairwise Welch t-tests** within that parameter:
 - Same Welch method as above.
 - **Pairwise multiple-testing correction (optional)** within that parameter's set of pairwise p-values: **FDR (BH)** or **Holm**.

All p-values are computed numerically via the regularized incomplete beta function (standard approach for t and F).

Output:

The header shows:

- Input file path.
- Detected groups (in order of first appearance).
- Your chosen omnibus and (if applicable) pairwise correction methods.
- Gatekeeping status (ON/OFF).
- Results table(s) per mode.



ANOVA across all samples

12. Apply ANOVA (contractions)

Two-group results:

Header columns include group names inside N and Mean labels, to avoid ambiguity:

Param	N1 (Cont)	N2 (H ₂ O ₂)	Mean1 (Cont)	Mean2 (H ₂ O ₂)	Diff(2-1)	t	DF	p (two-sided)	sig (p)
Loc. Maxamplitude [a.u]	4305	7821	6.628	7.158	0.53	5.434	9240.27	5.66E-08	****
Since last beat abs. time [sec]	4305	7821	0.992	0.983	-0.009	-1.061	9374.565	2.89E-01	ns
Peak time user absolute [sec]	4305	7821	0.17	0.164	-0.005	-1.687	9825.885	9.17E-02	ns
Contract. interval (user) absol. [sec]	4305	7821	0.06	0.062	0.002	0.799	11174.742	4.25E-01	ns
Relax. interval (user) absol. [sec]	4305	7821	0.11	0.103	-0.007	-3.851	8407.321	1.19E-04	***
Peak time Thr(10%) absol. [sec]	4305	7821	0.245	0.236	-0.01	-2.044	9695.98	4.10E-02	*
Contract. interval Thr(10%) absol. [sec]	4305	7821	0.08	0.083	0.003	1.13	10516.61	2.59E-01	ns
Relax. interval Thr(10%) absol. [sec]	4305	7821	0.166	0.152	-0.013	-5.067	8395.033	4.12E-07	****
Peak time Thr(50%) absol. [sec]	4305	7821	0.096	0.09	-0.006	-4.185	7436.399	2.89E-05	****
Contract. interval Thr(50%) absol. [sec]	4305	7821	0.039	0.038	-0.002	-2.243	8381.815	2.49E-02	*
Relax. interval Thr(50%) absol. [sec]	4305	7821	0.057	0.052	-0.005	-4.511	7161.614	6.54E-06	****
Peak time Thr(90%) absol. [sec]	4305	7821	0.033	0.032	-0.001	-4.204	8298.997	2.65E-05	****
Contract. interval Thr(90%) absol. [sec]	4305	7821	0.016	0.015	-0.001	-5.423	8574.514	6.02E-08	****
Relax. interval Thr(90%) absol. [sec]	4305	7821	0.019	0.018	-0.001	-4.363	8234.76	1.30E-05	****

N1/N2: sample sizes for each group for that parameter (rows with numeric data);

Mean1/Mean2: group means.

Diff(2-1): Mean(GroupB) - Mean(GroupA); **t, DF:** Welch t and degrees of freedom.

p(two-sided): p-value for the difference; **sig(p):** ns (not significant), * $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$, **** $p \leq 0.0001$.

If you chose an **omnibus correction**, you'll see an additional table listing **raw p** and **adjusted p** (q for FDR or pHolm) for each parameter.



ANOVA across all samples

12. Apply ANOVA (contractions)

≥3-group results (ANOVA):

Per parameter (in this example, 4 groups exist):

Param	k	N	DF1	DF2	F	p	sig (p)	eta^2
Loc. Maxamplitude [a.u]	4	19853	3	19849	112.75	2.15E-72	****	0.017
Since last beat abs. time [sec]	4	19853	3	19849	26.63	3.51E-17	****	0.004
Peak time user absolute [sec]	4	19853	3	19849	41.54	9.45E-27	****	0.006
Construct. interval (user) absol. [sec]	4	19853	3	19849	56.942	1.21E-36	****	0.009
Relax. interval (user) absol. [sec]	4	19853	3	19849	38.779	5.60E-25	****	0.006
Peak time Thr(10%) absol. [sec]	4	19853	3	19849	48.669	2.49E-31	****	0.007
Construct. interval Thr(10%) absol. [sec]	4	19853	3	19849	41.652	8.01E-27	****	0.006
Relax. interval Thr(10%) absol. [sec]	4	19853	3	19849	62.557	3.01E-40	****	0.009
Peak time Thr(50%) absol. [sec]	4	19853	3	19849	87.841	1.84E-56	****	0.013
Construct. interval Thr(50%) absol. [sec]	4	19853	3	19849	94.609	8.54E-61	****	0.014
Relax. interval Thr(50%) absol. [sec]	4	19853	3	19849	59.037	5.47E-38	****	0.009
Peak time Thr(90%) absol. [sec]	4	19853	3	19849	55.811	6.45E-36	****	0.008
Construct. interval Thr(90%) absol. [sec]	4	19853	3	19849	36.738	1.15E-23	****	0.006
Relax. interval Thr(90%) absol. [sec]	4	19853	3	19849	50.44	1.82E-32	****	0.008

k: number of groups; **N**: total valid rows for that parameter; **eta²**: proportion of variance explained by group (0–1);

Rough guide: ~0.01 small, ~0.06 medium, ~0.14 large (these are context-dependent).

If **omnibus correction** was selected, you'll see a table of **raw** and **adjusted** omnibus p-values across the 14 parameters.



ANOVA across all samples

12. Apply ANOVA (contractions)

If **gatekeeping** is ON, pairwise tables appear **only for parameters whose omnibus test passes $\alpha = 0.05$** (using corrected p if a correction was selected). Each parameter then lists its **pairwise Welch tests**:

Since last beat
abs. time [sec]

GroupA	GroupB	N1	N2	Mean1	Mean2	Diff(2-1)	t	DF	p	sig (p)	pHolm	sig(pHolm)
MaleCont	MaleNoise	6082	4216	0.861	0.84	-0.021	-3.444	8452.359	5.75E-04	***	1.73E-03	**
MaleCont	FemaleCont	6082	5649	0.861	0.826	-0.034	-7.055	11670.213	1.82E-12	****	9.10E-12	****
MaleCont	FemaleNoise	6082	3906	0.861	0.814	-0.046	-8.245	8595.036	1.90E-16	****	1.14E-15	****
MaleNoise	FemaleCont	4216	5649	0.84	0.826	-0.013	-2.334	7728.46	1.96E-02	*	3.92E-02	*
MaleNoise	FemaleNoise	4216	3906	0.84	0.814	-0.026	-3.974	8079.837	7.13E-05	****	2.85E-04	***
FemaleCont	FemaleNoise	5649	3906	0.826	0.814	-0.012	-2.248	7838.784	2.46E-02	*	3.92E-02	*

Files (if “Save results to folder” was set):

- **ENG file:** dot decimal.
- **GER file:** comma decimal.

The file content mirrors the console output, including correction summaries and (if applicable) pairwise sections.



ANOVA across all samples

12. Apply ANOVA (contractions)

Choosing corrections: practical guidance

Exploratory screens (you expect multiple parameters to move; you want sensitivity):

- **Omnibus:** FDR (BH) – keeps more power across the 14 parameter tests.
- **Pairwise ($k \geq 3$):** FDR (BH) – balances discovery and false positives within each parameter.

Confirmatory analyses (pre-registered hypotheses; strong control over any false positive):

- **Omnibus:** Holm – controls family-wise error across parameters.
- **Pairwise ($k \geq 3$):** Holm – strict per-parameter family-wise control over pairwise comparisons.

Gatekeeping ($k \geq 3$): ON

- Good default. It prevents pairwise testing when the omnibus isn't significant, reducing multiplicity and helping interpretation.



ANOVA across all samples

12. Apply ANOVA (contractions)

Interpreting common outcomes:

Significant omnibus p (after your chosen correction) with large η^2 :

Strong evidence that groups differ for that parameter, with a meaningful effect size.

→ Inspect pairwise table (if present) to see which groups differ.

Significant omnibus p but small η^2 :

Statistically detectable difference with small explained variance; may be biologically modest.

→ Still report; discuss practical relevance.

Omnibus not significant (gatekeeping ON):

No pairwise tests are shown for that parameter; interpretation is “no reliable group effect.”

If you must explore, disable gatekeeping but be transparent about multiplicity and adjusted p-values.

Two-group mode:

For each parameter, focus on **Diff(2-1)**, **p**, and consider presenting a CI (not in this macro) alongside stars to convey magnitude and uncertainty.



ANOVA across all samples

12. Apply ANOVA (contractions)

Troubleshooting & edge cases:

“Header should be: Group + 14 columns.”

Your file needs 15 tab-separated headers: 1 group name + 14 parameter names.

Strange Ns or missing lines:

Rows with non-numeric values in any parameter are skipped for that parameter. Check your file for stray text.

NaNs in output:

Occur when a group has $n < 2$ for a parameter, or when variance is undefined. Ensure enough observations per group.

Huge imbalance between groups:

Welch's tests are robust, but extreme imbalance reduces power. Consider reporting effect sizes and sample sizes clearly.



ANOVA across all samples

12. Apply ANOVA (contractions)

Recommended reporting template:

For **two groups** (per parameter):

“Welch t-test **comparing GroupB vs GroupA**: $\text{Diff}(2-1) = X$, $t(df) = T$, $p = P$ ($q = Q$ if FDR or $p\text{Holm} = H$ if Holm);
 $N(\text{GroupA})=N1$, $N(\text{GroupB})=N2$.”

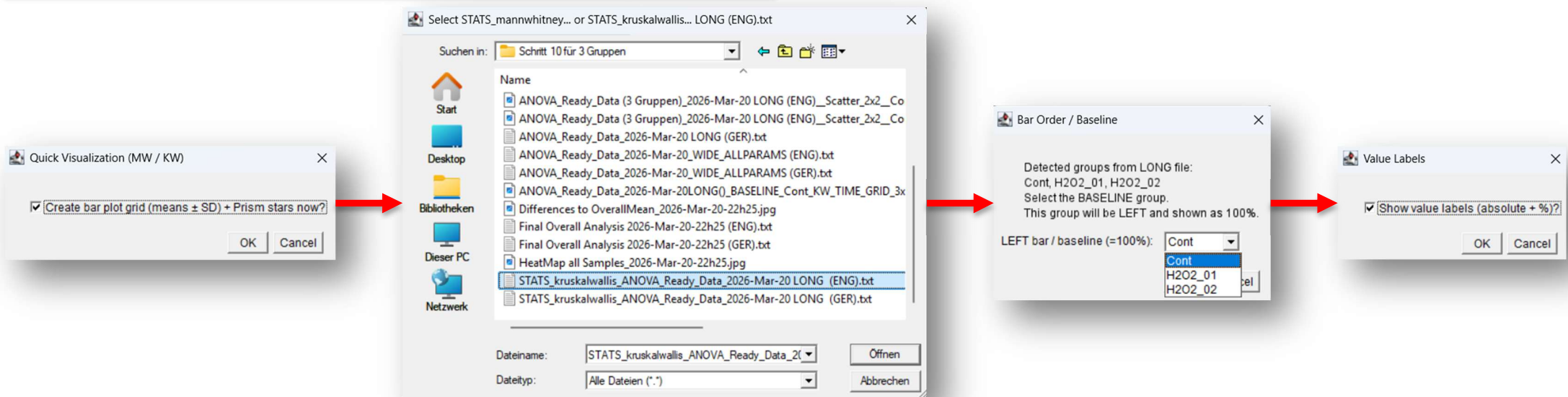
For **$k \geq 3$ groups** (per parameter):

“One-way ANOVA: $F(df1,df2) = F$, $p = P$ ($q/p\text{Holm}$ if corrected), $\eta^2 = E$. Pairwise Welch tests (gatekeeping **ON/OFF**): ...
(report adjusted p if used).”



ANOVA across all samples

13. Comparison & visualization (single cells)



1. You can skip the presentation of the mean values as a bar chart and proceed directly to the scatter plots (Step 5).

2. Select the “**STATS_mannwhitney LONG (ENG).txt**”-file (for 2 groups) or the “**STATS_kruskalwallis LONG (ENG).txt**” file (for 3 or more groups).

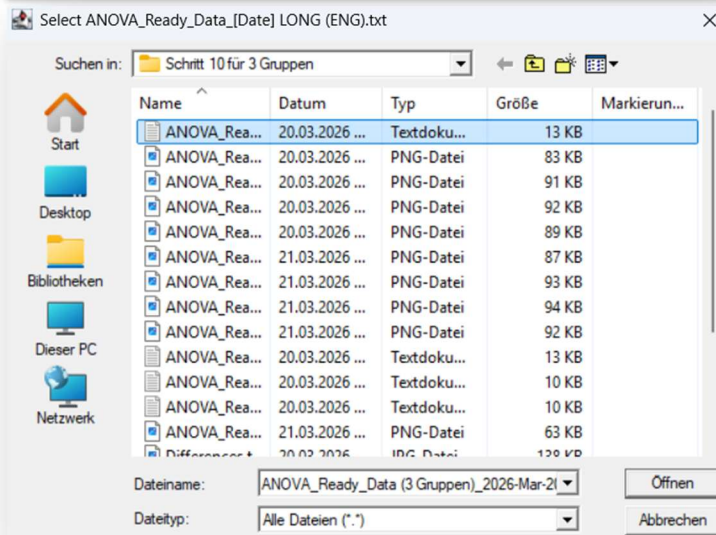
3. Define the group that should be set to 100% and displayed on the far left of the bar chart (typically the control group).

4. Specify the type of presentation (i.e., whether the numerical values should be displayed directly in the bar chart).

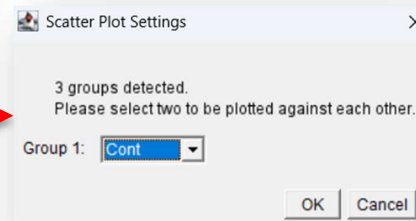


ANOVA across all samples

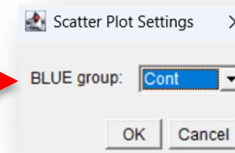
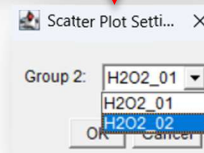
13. Comparison & visualization (single cells)



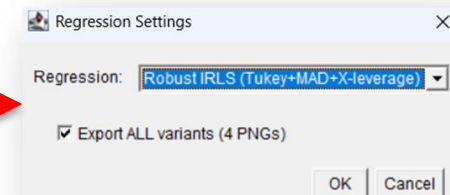
5. Select the “ANOVA_Ready LONG (ENG).txt”-file to create scatter plots.



6. Define which **two** groups should be plotted against each other.



7. Define which of the groups should be displayed in blue or red.

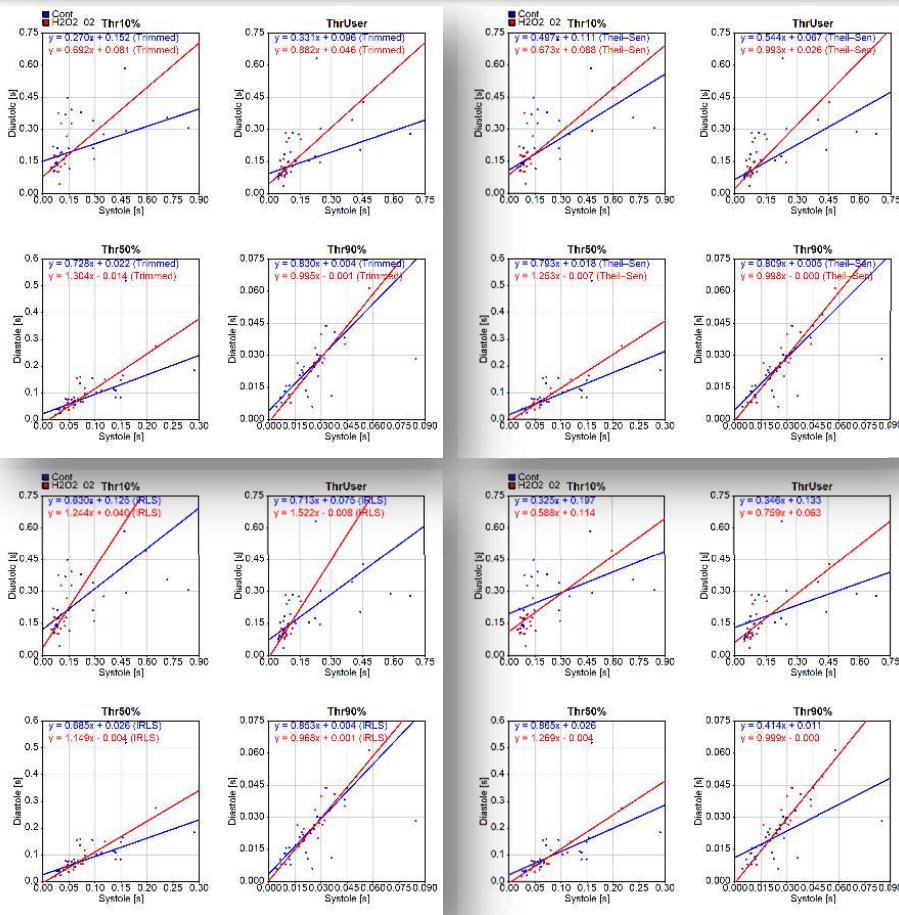


8. Define **which type of scatter plot** (or all types) should be created.

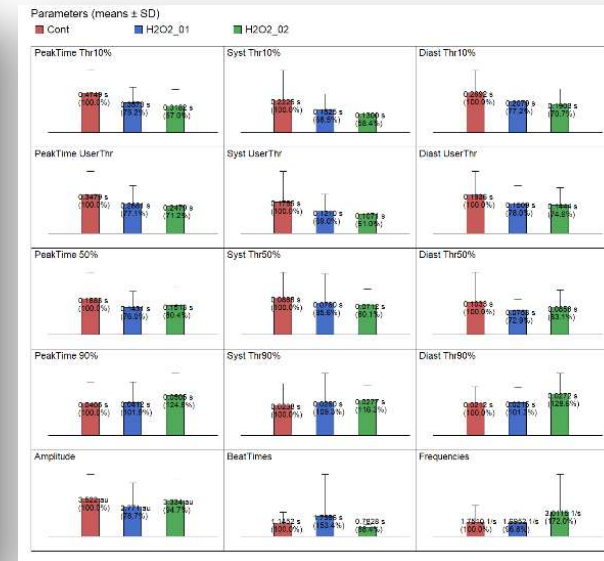
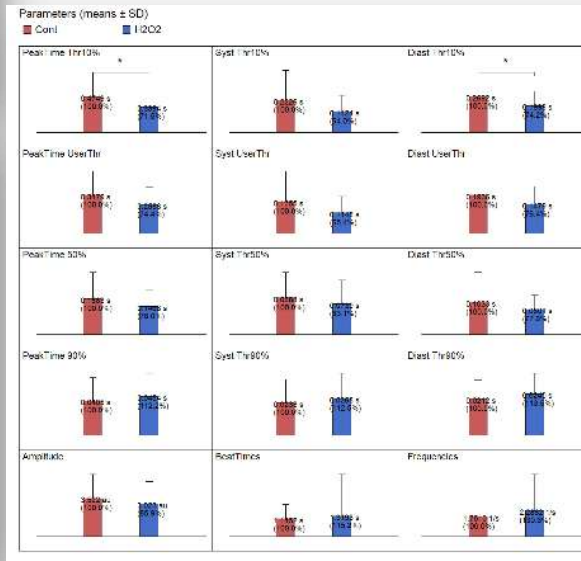


ANOVA across all samples

13. Comparison & visualization (single cells)



You will obtain a bar chart of your data (mean $\pm$ standard deviation, including the statistical significance from the previous analysis), as well as four different scatter plots (Theil-Sen, Trimmed, IRLS, and Classic) comparing two groups across all four thresholds.





ANOVA across all samples

14. Create motion-heatmaps

What it does

Turns each frame-difference video (“mjpeg-diff-cellavi”) into a single motion heatmap by summing “strong” frames and coloring the result (Fire/Thermal/Royal). You get quick, comparable images of where motion occurred across the whole clip.

How to run

- **Pick a folder (just select, do NOT open)**
 - If you **select a specific folder** named “diffMov” → **only that folder** is processed.
 - **Otherwise** select the **main folder** → the macro recursively finds and **processes ALL “diffMov”-subfolders**.
- **Set Cutoff (0-1); default set to 0.80.**
 - Frames (and also pixels) with mean intensity $\leq (\text{Cutoff} \times (\text{max frame mean}))$ are ignored before summation.

Attention: This process may take a (very) long time.



ANOVA across all samples

14. Create motion-heatmaps

Output

Per “diffMov” folder:

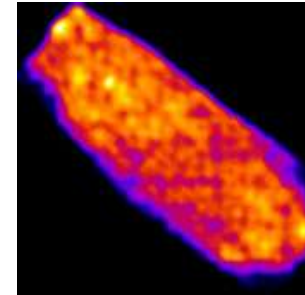
- **Per video:**
 - “Motion-Heatmaps/Fire/<name>_Fire.jpg”
 - “Motion-Heatmaps/Thermal/<name>_Thermal.jpg”
 - “Motion-Heatmaps/Royal/<name>_Royal.jpg”
- **Folder summaries (montages):**
 - “Motion-Heatmaps/Fire.jpg”, “Royal.jpg”, “Thermal.jpg”

How to read the image

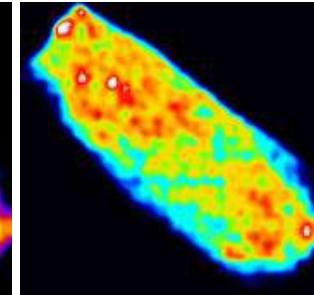
- Brighter = **more cumulative motion** over the whole video.
- Compare conditions by **extent**, **location**, and **intensity** of hot regions.

=> Also rapid identification of duplicate cells.

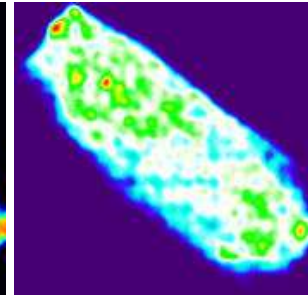
„Fire“



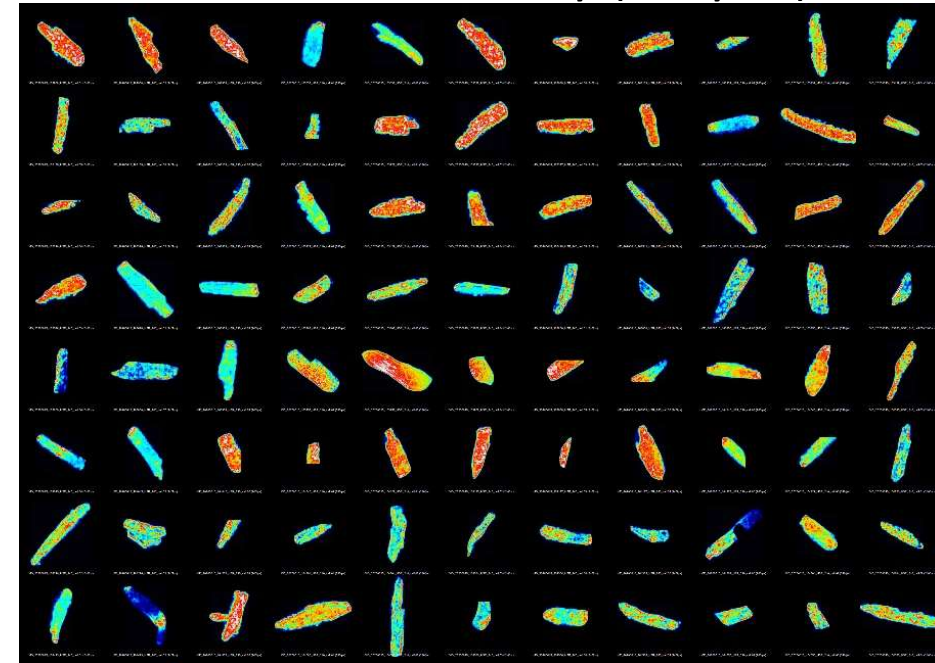
„Royal“



„Thermal“



Folder summary (“Royal”)





Files, output and downstream use

Step	Module	Required input	Generated output files	Used later in	User meaning / note
1	Pretest	Original AVI videos; LowMovement/HighMovement; optional first n frames only	pretest (<Setting>) [Date]/ <Video>-test.jpg	Step 2 (Batch List); visually also Step 4 settings	Choose threshold and size; no text data yet.
2	Create a Batch List	Pretest JPGs from Step 1; manual click selection	batch.txt	Step 4	Stores video-specific threshold/size values; copy into video folder before batch evaluation.
3	Manual ROI selection	Original AVI videos; manual ROIs; optional reference frame per cell	manrois [Date]/ roilist.txt diffroilist.txt ROI files (*.roi/*.zip)	Step 4	Optional alternative to automatic ROI detection.
4	Evaluation	AVI videos + threshold/size OR batch.txt OR manual ROI files	VidsMovement [Date]/ <Video>-cells.jpg <Video>-cell XX ([fps]fps).avi isolated/<Video>-diff-cell XX-isolated.avi diffMov/<Video>-diff-cell XX ([fps]fps).avi dataplots/Results [Date].txt dataplots/Amplitudes only [Date].txt dataplots/<Cell>-plot large/small.jpg optional: <Cell> animated (25fps).avi	Steps 5, 6, 7, 14	Central extraction step; Results and Amplitudes are the key data tables; diffMov feeds motion heatmaps.



Files, output and downstream use

Step	Module	Required input	Generated output files	Used later in	User meaning / note
5	Exclude Data	dataplots folder from Step 4, especially Results, Amplitudes only, and cell plots	Results (selected) [Date].txt Results (final summary) [Date].txt Amplitudes (selected) [Date].txt per-cell result and amplitude files	Steps 6, 7, 8, 9	Manual curation step. Use selected/final-summary files for most downstream analyses.
6	Re-Evaluation	Amplitudes only OR Amplitudes (selected)	<selected file> re-evaluated [Date].txt <selected cell> re-evaluated N [Date].jpg	—	Optional side branch; re-evaluated outputs are not automatically collected by later steps.
7	Data Extraction	Results file, preferably Results (selected)	Extracted results (eng) <Results filename>.txt Extracted results (ger) <Results filename>.txt Extracted results Plot <filename>.jpg Histogram *.jpg Scatter plots *.jpg CorrHeatmap_<filename>.jpg	Step 11	Contraction/event-level table. ENG file is the important downstream file.
8	Extraction for Statistics	Results (final summary) from Step 5	<MainFolder>_Results Final Summary for Stats (ENG) [Date].txt <MainFolder>_Results Final Summary for Stats (GER) [Date].txt	Step 10	Condenses curated cell summaries into sample-level stats input.



Files, output and downstream use

Step	Module	Required input	Generated output files	Used later in	User meaning / note
9	Statistics for selected Samples	Results (selected) from Step 5	Statistics_[Date_Time]/ <Cell>_Statistics.jpg *_ANOVA-Heatmap.jpg _ANOVA-All Parameters.jpg Results Final Summary for Stats [Date] (ENG/GER).txt Results ANOVA all Cells [Date] (ENG/GER).txt	Step 10	Within-sample/cell-to-cell comparison; can also produce files Step 10 collects.



Files, output and downstream use

Step	Module	Required input	Generated output files	Used later in	User meaning / note
10	Overall Statistics (for cells)	Root folder containing one or more *_Results Final Summary for Stats (ENG).txt files from Step 8 and/or 9	ANOVA_Ready_Data_[Date] LONG (ENG/GER).txt ANOVA_Ready_Data_[Date]_WIDE_ALLPARAMS (ENG/GER).txt optional: ANOVA_Ready_Data_[Date] EXCLUDED_INVALID_ROWS (ENG/GER).txt optional: STATS_mannwhitney_* OR STATS_kruskalwallis_* (ENG/GER).txt HeatMap all Samples_*.jpg Differences to OverallMean_*.jpg Final Overall Analysis * (ENG/GER).txt Normalization_REFERENCES_ControlMedian_[Date] (ENG/GER).txt Normalized_RATIO_raw_div_ControlMedian_BLOCKWID E_ControlLeft_TreatedRight_[Date] (ENG/GER).txt Normalized_LOG2_raw_div_ControlMedian_BLOCKWID E_ControlLeft_TreatedRight_[Date] (ENG/GER).txt Normalization_WARNINGS_ControlMedian_[Date] (ENG/GER).txt	Step 13 for LONG/STATS; normalized BLOCK-WIDE mainly for external analysis/export	Main cell-level comparison. Current normalization writes BLOCK-WIDE normalized files only, not normalized LONG files.



Files, output and downstream use

Step	Module	Required input	Generated output files	Used later in	User meaning / note
11	Prepare data for ANOVA (contractions)	Root folder with Extracted results (eng) files from Step 7	FOUND_files.txt (optional) Group_<Name>_merged (ENG/GER).txt ALL_groups_MERGED (ENG/GER).txt	Step 12	Merges contraction-level files by group.
12	Apply ANOVA (contractions)	ALL_groups_MERGED (ENG).txt	STATS_ttest (ENG/GER).txt OR STATS_anova (ENG/GER).txt	—	Final statistics for contraction-level branch; not the input for Step 13.
13	Comparison & visualization (single cells)	For bar plots: STATS_mannwhitney_* or STATS_kruskalwallis_* from Step 10; for scatter plots: ANOVA_Ready_Data_[Date] LONG (ENG).txt	<STATS base>_BASELINE_<Group>_<MW/KW>_TIME_GRID_3 x5.png <ANOVA_Ready_Data...>__Scatter_2x2__<Groups>__ Classic.png ...__IRLS_Tukey.png ...__TheilSen.png ...__Trimmed80.png	—	Final visualization of cell-level/sample statistics.



Files, output and downstream use

Step	Module	Required input	Generated output files	Used later in	User meaning / note
14	Create Motion-Heatmaps	diffMov folder or root folder containing diffMov folders from Step 4	Motion-Heatmaps/Fire/*.jpg Motion-Heatmaps/Thermal/*.jpg Motion-Heatmaps/Royal/*.jpg Motion-Heatmaps/Fire.jpg, Thermal.jpg, Royal.jpg Heatmaps Intensity (ENG/GER).txt Heatmaps Intensity UNITED (ENG/GER).txt Summary_Analysis_Plots.jpg	—	Final motion-analysis branch based on frame-difference movies.



Troubleshooting – Too little RAM

A **typical problem** is the premature abort of video analysis due to a too **small working memory**, caused by **too high resolution and/or too long videos**.

To analyze such videos nevertheless, their **spatial resolution CAN be reduced** (details on the next page).

However, the **temporal resolution** of the videos to be analyzed **should NEVER(!) be reduced**.

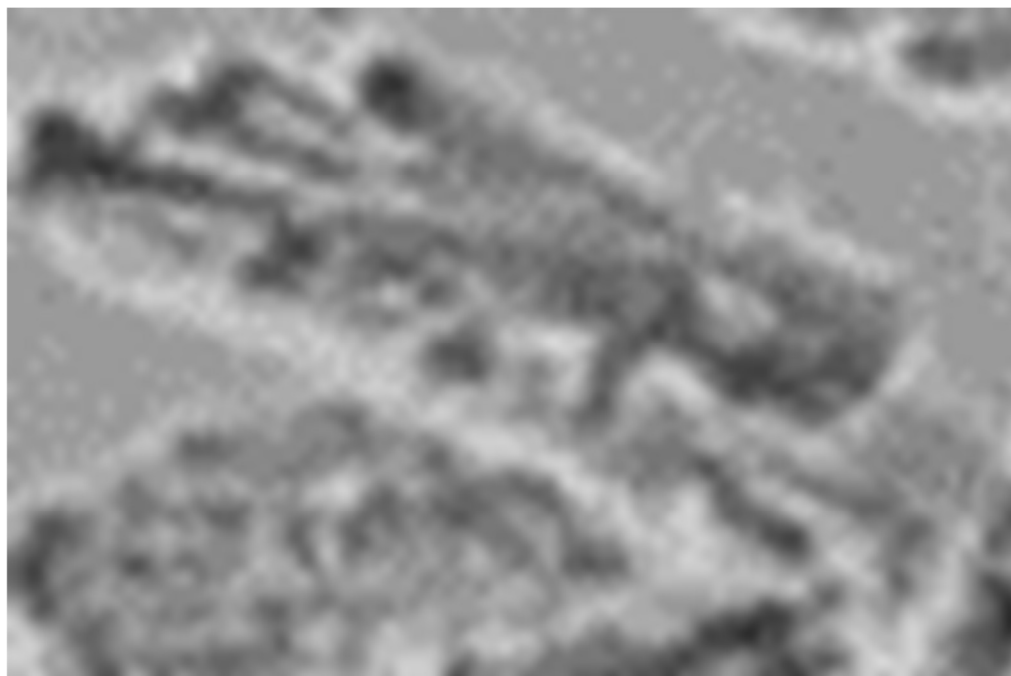


Troubleshooting – Too little RAM

Data reduction of **spatial resolution**:

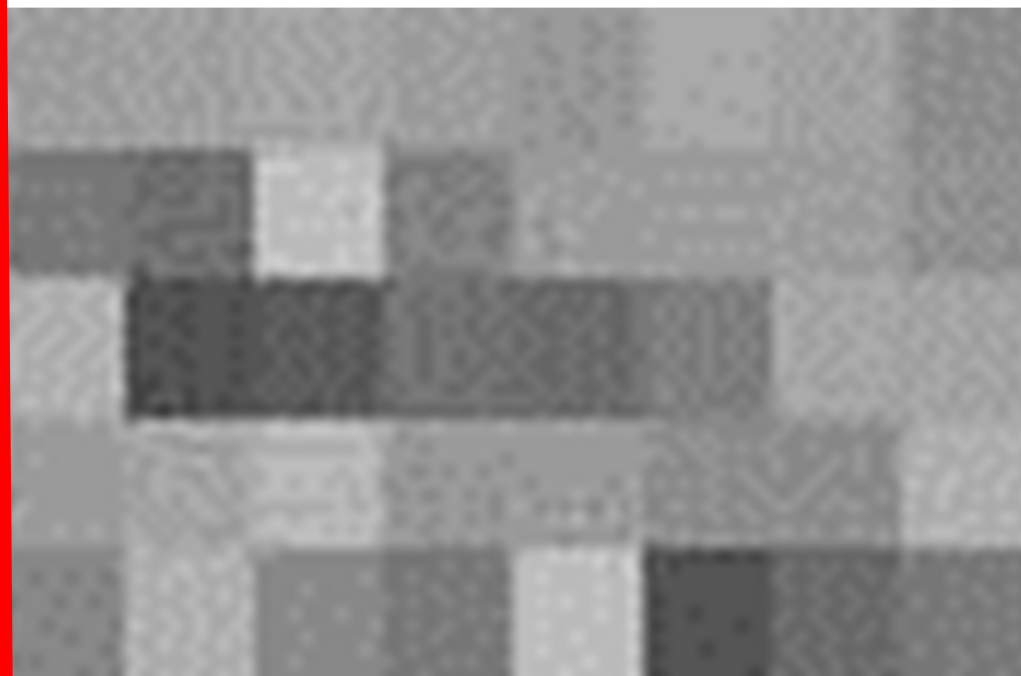
Original video (representative material):

87 x 57 = 4959 pixels (**100%**),
21,2 MB; 4300 single frames



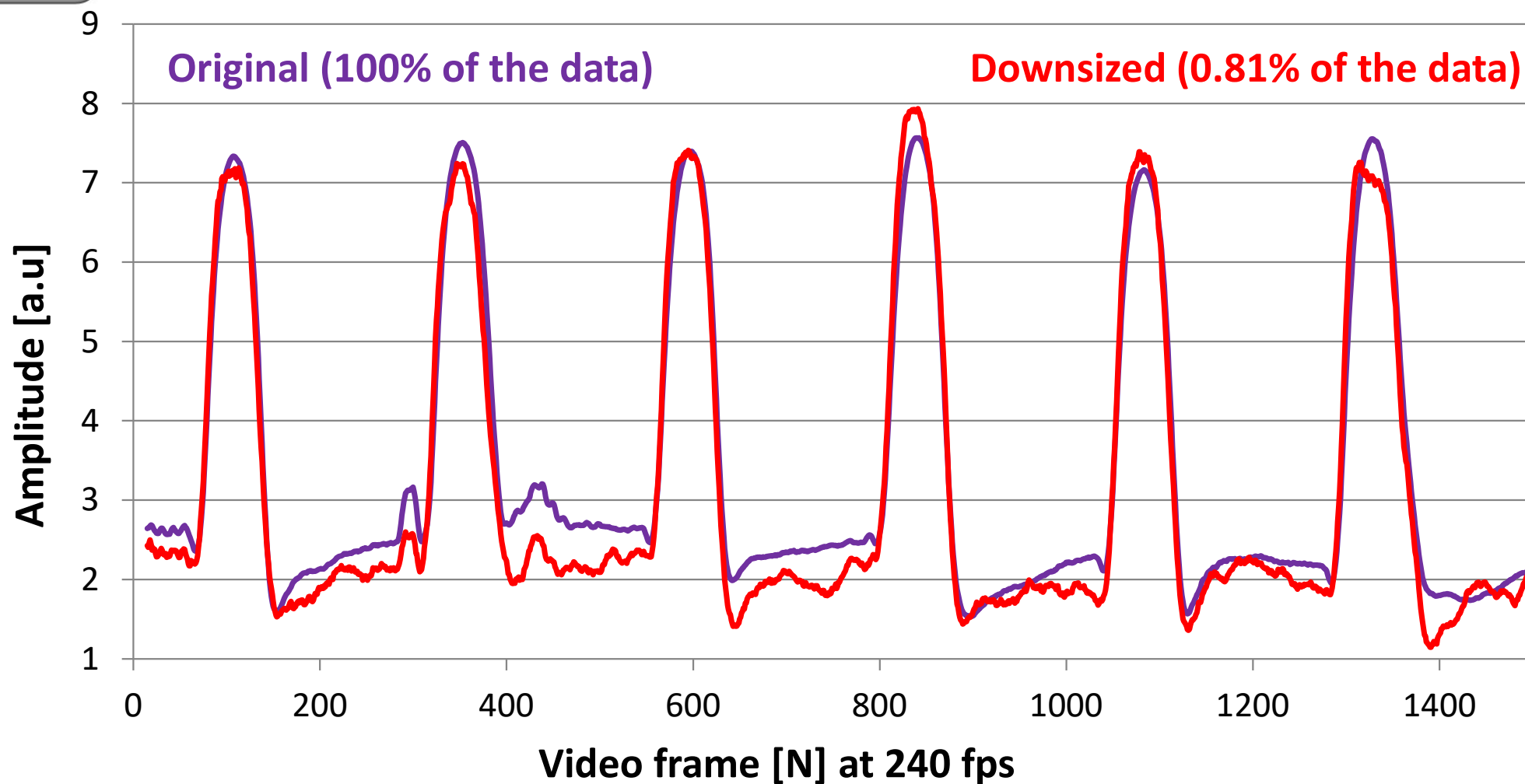
Downsized version (hardly recognizable):

8 x 5 = 40 pixels (**≈0.81%**),
0,273 MB; 4300 single frames





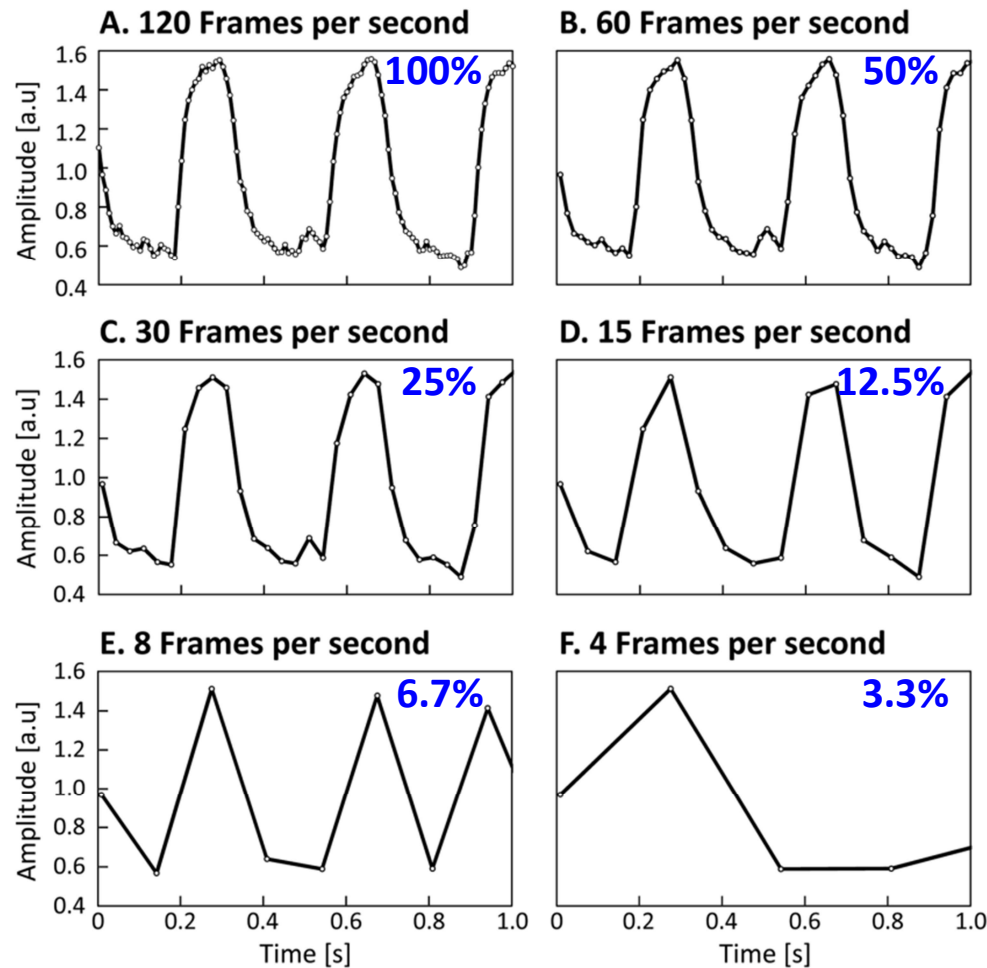
Troubleshooting – Too little RAM





Troubleshooting – Too little RAM

Data reduction of temporal resolution:



The importance of a high frame rate for successful evaluation of cardiomyocyte videos has already been discussed in the “MYOCYTER” publication*.

Recommended minimum: 120 fps, we are already using 240 fps.

Below 25% / 30 fps the videos are practically no longer evaluable

A too low frame rate of the videos to be evaluated is a devastating problem that cannot be solved afterwards.

* The „MYOCYTER” - Convert cellular and cardiac contractions into numbers with ImageJ. Grune, T.; Ott, C.; Häseli, S.; Höhn, A.; Jung, T. *Sci Rep.*, **9(1):15112**, 2019.



Troubleshooting – Too little RAM

In sum:

While **temporal compression quickly results in a disaster** (below 30 fps for cardiomyocytes, 25% of the recommended minimum frame rate of 120 fps),

spatial compression to $\approx 1\%$ of the original resolution is **actually acceptable**.



Please cite us when you use MYOCYTER

If you use our macro, **please cite** the corresponding **peer-reviewed publication**:

**“The MYOCYTER – Convert cellular and cardiac contractions into numbers with ImageJ”,
Grune, T.; Ott, C.; Häseli, S.; Höhn, A.; Jung, T., *Scientific Reports*, October 2019.**

or

**“MyoCyter v2.0: an open-source ImageJ macro for cardiomyocyte contractility analysis”,
Scientific Reports, 2026.**

If you have **any questions about the software**, please contact me:

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